Opacity as Commitment: Fragile Equilibria under Delegated Trading*

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January 18, 2026

Abstract

We study a model of delegated asset management in which a fund manager care not only about trading profits but also her reputation, i.e., clients' perception of the manager's skill. Fund opacity, defined as imperfect information about the fund's strategy, generates two self-fulfilling equilibria: a benchmark and a "flow-driven" equilibrium. In the benchmark equilibrium, the manager prioritize the fund performance, and her trading intensity is at the profit-maximizing level. In the latter, however, the fund manager trades more aggressively to appear skilled, thereby attracting larger fund flow and associated fund management fees. The flow-driven equilibrium features poorer performance and higher price volatility. We show that (1) labor mobility in finance generates traders' reputation concerns, triggering market fragility and increasing wage inequality, (2) traders with different skill levels adopt different trading styles, and (3) skills in active management may not translate into performance.

*For helpful comments and discussions, we thank Zhi Da, Paul Ehling, Albert Kyle, Yasuto Monden, Don Noh, Yoshihiro Ohashi, Maureen O'Hara, Wataru Ohta, Norman Schü, Yung Chiang Yang, and participants at Economic Theory and Policy Workshop (Tokyo), Hitotsubashi University, Nippon Finance Association, Osaka University, SFI Research Days, Summer Workshop on Economic Theory (SWET), Tokyo Metropolitan University, Asian Finance Association Conference, University of Lausanne, and HKUST. This work is supported by JSPS KAKENHI Grant Number JP17H07084. The views and conclusions expressed in this paper are strictly those of the authors. They do not necessarily reflect those of Peritus Investment Consultancy. Peritus Investment Consultancy takes no responsibility for any errors or omissions in the information contained in this paper. All errors and omissions are our own.

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JEL classification: G11, G12, G14, G23

Key words: informed trading, market reputation, trader skill, financial fragility, poaching, asset prices

Conflict-of-interest disclosure statement

Jun Aoyagi: I have nothing to disclose.

Yuki Sato: This work was supported by KAKENHI (Grants-in-Aid for Scientific Research) of JSPS (Japan Society for the Promotion of Science), FY 2017-2019. The Grant Number is JP17H07084, and the total amount is JPY 2,100,000 (about USD 14,000).

Raluca Toma: I have nothing to disclose.

1 Introduction

Financial firms compete for skilled employees. As the boundaries between investment banking, commercial banking, and asset management blur, this competition intensifies. Following the 2007-2009 crisis, pressures to reduce compensation and the resulting challenges in retaining talent have brought poaching practices into focus.[1] Poaching, or more generally labor mobility, raises concerns among financial specialists about their reputation within the industry. How do these concerns influence their trading strategies and skill acquisition? How do they interact with asset prices and market liquidity?

To address these questions, we study a model inspired by Kyle (1985), where traders, privately informed about their skill, trade risky assets anonymously with market makers. We define skill as the trader-specific ability to uncover more information about asset payoffs than the market. The more skilled a trader is, the larger the potential mispricing she can identify, and thus, the greater the trading profits she can potentially earn. Each trader selects an asset based on her skill and strategically submits a market order to the market maker for that asset, who sets the price after observing the aggregate order flow. A recruiter attempts to infer each trader's skill from the observed price and payoff of the asset chosen by the trader. The trader values not only trading profits but also her reputation, i.e., the recruiter's estimate of her skill. In a simple one-period model presented in Section 2, each trader derives an exogenous benefit from her reputation, which we model as an increasing quadratic function of reputation. In Section **??**, we extend the model to a two-period setting, providing a microfoundation for the reputation benefit. In this extension, each trader randomly encounters a recruiter who compensates her for her skill based on his estimate. In equilibrium, the quadratic form

[1] According to *The Wall Street Journal*, poaching is a common practice in the sophisticated capital markets industry, occurring in both bull and bear markets ("Greed is Good," February 2009), and employees at all levels are being poached ("Employee Poaching Returns to Wall Street," May 2010). *The New York Times* ("A Mad Scramble for Young Bankers," July 2014) and *Bloomberg* ("Saturdays Off Lose to Cash as Buyout Firms Poach Junior Bankers," March 2014) highlight the buy-side's (private equity, hedge funds) recruitment of young talent from investment banking amid a changing financial industry.

assumed in Section 2 arises endogenously.

The model yields two stable, self-fulfilling equilibria, one of which is similar to that of Kyle (1985), where reputation is irrelevant. In the other, the "reputation-driven" equilibrium, traders trade more aggressively to appear skilled. This strategy increases trading volume, price volatility, and market liquidity but undermines trading profits by revealing too much private information to the market maker. Intuitively, each trader can boost her reputation by placing a large order, either buy or sell, because such an action makes her appear like a skilled trader who has identified an asset with significant mispricing. In equilibrium, the recruiter accurately anticipates the trader's strategy, so the recruiter's learning is not manipulated; nevertheless, the trader may trade aggressively depending on the market's belief. To illustrate the mechanism, suppose the recruiter believes that the trader trades more aggressively than the profit-maximizing level. If she trades less aggressively to increase profits, the recruiter (incorrectly) infers that she is less skilled. To avoid the reduction in her reputation benefit, she conforms to the recruiter's belief and indeed trades very aggressively.

The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the reputation-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.[2] Key to our multiple equilibria and fragility results are the trader's conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading profits, but on the other hand, she wants to show it off to the recruiter to obtain the reputation benefit. If the trader's concern for reputation is very weak (very strong), the first (second) motive dominates, and the Kyle-like (reputation-driven) equilibrium holds uniquely. However, if the concern is moderately strong, either equilibrium may arise depending on the market's self-fulfilling beliefs. Therefore, the model predicts that fragility may occur

[2]As in Greenwood and Thesmar (2011), we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

when poaching is (moderately) active and there is high demand for financial skill, as these situations increase the likelihood of future recruitment and lead to heightened reputation concerns.

Our results shed light on how traders' underlying skills relate to their trading styles and contribute to financial fragility. The model predicts that high-skilled traders adopt low-intensity, profit-driven strategies, while those with lower skills engage in aggressive, risk-seeking behavior to gain reputation. Traders with intermediate skill levels, facing multiple equilibria, may switch between profit-seeking and reputation-driven strategies based on market beliefs, thereby contributing to market fragility. This behavior aligns with real-world observations during financial crises, such as momentum crashes and sudden reversals, which occur even without fundamental shocks.

The model has implications for compensation in the finance industry. In the reputation-driven equilibrium, traders' wages exhibit greater inequality among high- and low-skilled traders compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skill more than they do in the Kyle-like equilibrium. This allows the recruiter to better assess each trader's skill and, therefore, to offer wages that more accurately reflect their abilities. This result is consistent with the findings by Philippon and Reshef (2012) and Célérier and Vallée (2019), who document increasing returns to talent and their significant contribution to income inequality in the finance industry.

The aggressive trade in the reputation-driven equilibrium also causes traders to reveal excessive information to the market maker, resulting in poorer trading performance than in the Kyle-like equilibrium. Consequently, high-skilled traders in the reputation-driven equilibrium may underperform low-skilled traders following the Kyle-like equilibrium. This result offers a novel explanation for the puzzling relationship (or the lack thereof) between trader skill and persistent performance, as documented and analyzed in the literature (e.g., Busse, Goyal, and Wahal, 2010; Berk and Green, 2004).

In Section **??**, we study traders' endogenous skill acquisition. We show that, due to the multiplicity of equilibria, each trader may intentionally acquire a “middling” level of skill, which is too high or too low compared to the level she would acquire if she knew which type of equilibrium would prevail. Ex-post, in the Kyle-like equilibrium, where profits are prioritized over reputation, the acquired skill level is insufficient. Conversely, in the reputation-driven equilibrium, where reputation outweighs profit motives, the skill level is excessive. Ex-ante, without knowing which equilibrium will emerge, the trader optimally chooses a middling skill level to prepare for both scenarios. This finding aligns with real-world observations that some finance professionals tend to pursue broader business qualifications rather than specializing in areas like advanced computer science or pursuing higher academic degrees such as a PhD.

Our paper is related to the literature on fragility in financial markets. Cespa and Vives (2015) is most closely related, as they also find multiple equilibria in an asymmetric information trading model. The key ingredients of their model are short-term investor horizons and persistent liquidity trading, which generate strategic complementarities. In Froot, Scharfstein, and Stein (1992), strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from traders’ forward-looking concerns about future recruitment opportunities. In Cespa and Foucault (2014), learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks.

Our paper also connects to the literature on trading frenzies, such as Veldkamp (2006) and Goldstein, Ozdenoren, and Yuan (2013), in which strategic complementarities play a key role. In Veldkamp (2006) frenzies originate from complementarities in information markets, while in Goldstein, Ozdenoren, and Yuan (2013) strategic complementaries, triggered by a feedback effect from financial markets to a firm’s real investment, lead to coordinated aggressive trading. By contrast, in our reputation-driven equilibrium, each trader trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. DiMaggio (2015), Malliaris and Yan (2021), and Bijlsma, Boone, and Zwart (2018) derive implications for the risk-taking behavior of agents when ability is private information but is reflected in performance. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications. Guerrieri and Kondor (2012) show that career concerns amplify price volatility, similar to our findings. In Dasgupta (2006), career concerns lead to churning (trading without private information), and in Dasgupta and Prat (2008), they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility, which is the main focus of our paper.

The rest of the paper proceeds as follows. Section 2 presents a baseline model with exogenous reputation concerns, and Section 3 studies the equilibrium. Section **??** endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section **??** analyzes traders' skill acquisition. The Online Appendix contains all proofs for the theoretical results.

2 Model

This section presents a one-period trading model à la Kyle (1985) with delegated asset management. The economy consists of a *fund manager*, a competitive *market maker*, a *noise trader*, and $n \geq 2$ *fund clients (investors)* indexed by $i = 1, 2, \cdots, n$. All players are risk neutral. The manager leverages her skill (defined later) to identify a single risky asset with potential mispricing, raises capital from clients, and allocates it between the risky asset and a risk-free asset with a zero interest rate.[3]

[3]We abstract away from information acquisition by individual clients and assume that they invest through a manager due to the lack of informational advantages. This assumption

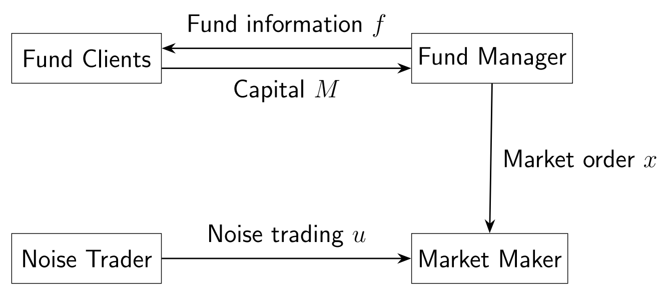


Figure 1: Capital Allocation and Investment

Skill. The manager can uncover more information about the asset's payoff than the market maker, and we interpret it as her *skill*. At the beginning of the period, the manager draws private information s from a normal distribution with mean zero and variance ω_s. Leveraging this information, she identifies an asset with payoff $\delta = \bar{\delta} + s$, where $\bar{\delta}$ is a publicly know mean payoff. Since the market maker does not observe s, her prior expectation of the asset's payoff is $\mathrm{E}[\delta] = \bar{\delta} + \mathrm{E}[s] = \bar{\delta}$, which is biased by s from the informed manager's perspective. As the market maker would set a price $p = \mathrm{E}[\delta] = \bar{\delta}$ in the absence of traders, s can be viewed as the asset's "potential mispricing."[4] Indeed, as shown later, the manager profits by buying the asset if $s > 0$ (underpriced) and selling it if $s < 0$ (overpriced); the larger the $|s|$, the larger the profit. In light of this, we measure the manager's asset-selection skill by $|s|$.[5]

Investment. We model the manager's capital-raising and trading decisions as a simultaneous game with rational expectations, as illustrated in Figure 1. On the one hand, the manager chooses the allocation between the risky and the safe asset. This action is represented by a market order for $x \in$

would be supported by, for example, information- or skill-acquisition costs, where only the fund manager acquires the skill at lower costs due to economies of scale (e.g., Gârleanu and Pedersen, 2018).

[4] s is not the *actual* mispricing because the market maker will take into consideration the presence of the informed manager; thus, p will deviate from $\bar{\delta}$ in equilibrium.

[5] This formalization of skill aligns with findings in the empirical studies that attribute fund managers' performance to stock-picking skills (e.g., Wermers, 2000; Kosowski, Timmermann, Wermers, and White, 2006).

$\mathbb{R}$ units of the risky asset, submitted to the market maker in the financial market. The manager decides on her investment strategy by incorporating the reactions of the financial market (i.e., the asset's price) and the capital inflow from her clients. On the other hand, each client chooses how much capital to allocate to the fund manager based on *fund information*. In particular, the manager disseminates a noisy signal about her trading strategy to clients, denoted by $f = x + e$, where e represents a normally distributed noise term with mean zero and variance ω_e.[6] This signal is conveyed through private interactions between the manager and her clients and can be interpreted as the fund manager's practice of communicating her strategy, such as a high-conviction or concentrated risky position, to clients in order to attract capital.[7] Since the manager chooses her trading strategy x based on the private signal s, the fund information f is informative about the manager's skill and the fund's return. Thus, fund clients use f to decide their capital allocations. We denote client i's capital allocation to the fund as $m_i \geq 0$, and the fund's total assets under management (AUM) as $M \equiv \sum_{i=1}^{n} m_i$. The fund's fee structure is proportional to total profits: the manager takes a $\phi \in (0, 1)$ fraction of the total fund profits, while the remaining $1 - \phi$ fraction is distributed to the clients in proportion to their contributions to the fund, i.e., m_i/M.[8]

Financial market. In the financial market, the market maker sets the asset price, p, based on the incoming order flow, $x + u$, where u represents a random market order from the noise trader and follows a normal distribution with mean zero and variance ω_u. Due to competition, the market maker sets a

[6] The fund manager may deviate *ex post* from the strategy initially communicated to clients. Such incentives, as well as the role of monitoring in preventing deviations, are analyzed in Section 5.

[7] Such activities are consistent with the empirical evidence and theoretical rationale in Anton, Cohen, and Polk (2021) and Van Nieuwerburgh and Veldkamp (2010).

[8] Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists, $0 < \phi < 1$.

semi-strong efficient price conditional on $x+u$:[9]

$$p = \mathrm{E}[\delta | x+u]. \tag{2.1}$$

Fund opacity. The noise variance of the fund information, $\mathrm{Var}[e] = \omega_e$, is interpreted as *opacity* of the fund. It could arise from strategic obfuscation, whereby the fund deliberately increases its complexity and obscures the investment strategy to hinder uninformed clients from accurately inferring its true actions. In what follows, we first consider the baseline model with a perfectly transparent fund ($\omega_s = 0$), while Section 4 deals with general opacity ($\omega_s > 0$). Section 5 considers the manager's strategic choice of fund opacity, where the number of clients n also endogenously responds to ω_e. It uncovers the cost and benefit of an opaque fund, providing implications for equilibrium investment strategies and financial market quality.

Maximization problem. The fund's profit from the risky investment is represented by $\pi \equiv (\delta - p)x$, so that the total fund proceeds, including those from the safe asset, are

$$y = \pi + M. \tag{2.2}$$

Given the realized skill s, the fund manager maximizes her expected fee income by controlling her investment strategy x:

$$\max_x \mathrm{E}\left[\phi y | s\right]. \tag{2.3}$$

Conversely, conditional on the fund information f, client i maximizes her expected return from capital allocation to the fund by choosing m_i:

$$\max_{m_i} \mathrm{E}\left[\frac{m_i}{M}(1-\phi)y - m_i | f\right]. \tag{2.4}$$

Definition 1. *The equilibrium is defined by the market maker's pricing strat-*

[9] Assume that, on the equilibrium path, one competitive market maker trades the asset. As in Kyle (1985), competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

egy p, the fund manager's trading strategy x, and clients' capital allocations $\{m_i\}_{i=1}^n$, such that, (i) p satisfies (2.1) given $x+u$, (ii) x solves (2.3) given s, p, and $\{m_i\}_{i=1}^n$, and (iii) for each $i = 1, \cdots, n$, m_i solves (2.4) given p, f, and $\{m_j\}_{j\neq i}$.

3 Equilibrium

In this section, we search for a linear equilibrium under transparency ($\omega_e = 0$), in which the manager's market order x is linear in her skill s, and the market maker's pricing strategy p is also linear in the order flow $x + u$. Proofs for the analytical results are provided in the Appendix.

3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \tag{3.1}$$

where $\beta > 0$ is an endogenous constant determined in the equilibrium. Conjecture (3.1) states that the manager buys (sells) the asset if her s is positive (negative), where the order size is proportional to her skill $|s|$; the more skilled the manager, the more aggressively she trades.[10] The scale factor β is referred to as the *trading intensity*, representing how intensively the manager exploits her informational advantage over the market maker.

3.2 Execution Price

The market maker decides on the price of the asset by extracting information about the asset's payoff from the aggregate order flow $x + u$. Anticipating the

[10]This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e., $\mathrm{E}[\delta|s] - \mathrm{E}[\delta] = s$.

trading strategy (3.1), the price in equation (2.1) can be rewritten as

$$p = \bar{\delta} + \lambda(x + u), \tag{3.2}$$

where

$$\lambda \equiv \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u} \tag{3.3}$$

is referred to as "Kyle's lambda," a measure of the price impact of order flow.[11] Following the literature, we define a market to be *liquid* if λ is small.

3.3 Clients' Capital Allocation

By incorporating the fund proceeds $y = \pi + M$, client i's optimization problem in (2.4) is

$$\max_{m_i} \frac{m_i}{M}(1 - \phi)\mathrm{E}\left[\pi|f\right] - \phi m_i. \tag{3.4}$$

The cost of delegating capital management arises from the fee payment and is linear in the dollar value of the capital supply, as represented by ϕm_i in the second term of (3.4). In contrast, the marginal benefit is diminishing in m_i, as the fund's proceeds are allocated to clients on a pro-rata basis. This structure ensures the second-order condition of the problem (3.4), and the first-order condition leads to the optimal capital allocation as follows:

$$m_i = \left(\frac{1 - \phi}{\phi}\mathrm{E}[\pi|f]\sum_{j \neq i} m_j\right)^{\frac{1}{2}} - \sum_{j \neq i} m_j. \tag{3.5}$$

We focus on the symmetric equilibrium where all clients provide the same amount of capital to the fund, i.e., $m_i = m$ for all $i = 1, 2, \cdots, n$. In this

[11] From (2.1), we have $p = \mathrm{E}[\delta|x + u] = \mathrm{E}[\mathrm{E}[\delta|x + u, s]|x + u] = \mathrm{E}[\bar{\delta} + s|x + u]$. Since the market maker believes that the manager follows (3.1), s and $x + u$ are both normally distributed from the market maker's perspective. Thus, $p = \bar{\delta} + \mathrm{E}[s] + \frac{\mathrm{Cov}[s, x+u]}{\mathrm{Var}[x+u]}(x + u - \mathrm{E}[x + u]) = \bar{\delta} + \frac{\mathrm{Cov}[s, \beta s+u]}{\mathrm{Var}[\beta s+u]}(x + u - \mathrm{E}[\beta s + u]) = \bar{\delta} + \frac{\beta\omega_s}{\beta^2\omega_s+\omega_u}(x + u)$.

equilibrium, equation (3.5) is simplified to

$$m = \frac{n-1}{n^2}\frac{1-\phi}{\phi}\mathrm{E}[\pi|f]. \tag{3.6}$$

It suggests that the AUM of the fund is

$$M \equiv nm = \frac{n-1}{n}\frac{1-\phi}{\phi}\mathrm{E}[\pi|f]. \tag{3.7}$$

Individual and total fund contributions are proportional to the expected trading profit conditional on the fund information, $\mathrm{E}[\pi|f]$. This naturally follows from the performance-based fee structure, which makes the per-unit return from delegated asset management proportional to the trading profit. Equation (3.7) defines the coefficient of fund flow as

$$\theta \equiv \frac{n-1}{n}\frac{1-\phi}{\phi}. \tag{3.8}$$

Parameter θ is referred to as the *flow-performance sensitivity* in the following discussion. It increases with the number of clients (n), as a larger pool of clients contribute to the total delegation despite smaller individual supply, while it decreases with the fee rate (ϕ), as it raises the cost of delegation and discourages investment.

To compute the expected fund profits, each client seeks to back out the realized skill level of the manager $|s|$ from fund information $f = x + e$. When the fund information is perfectly transparent (i.e., $\omega_e = 0$), clients rationally use the trading strategy in (3.1) and infer $s = x/\beta$, leading to

$$\mathrm{E}\left[\pi|f\right] = (1-\lambda\beta)\beta\mathrm{E}[s^2|s = x/\beta] = \left(\frac{1}{\beta} - \lambda\right)x^2. \tag{3.9}$$

The first equality in (3.9) suggests that the fund return, $\pi = (\delta - p)x$, becomes a quadratic function of s, as both the manager's trading quantity x and the profit margin, $\delta - p$, are proportional to the skill level in expectation. Furtheremore, the transparent fund information indicates that $s = x/\beta$, thereby shaping the conditional expected profit into quadratic in x (the second equality in [3.9]). This result implies that, on top of the maximizing profits, delegated asset management generates an additional incentive for the manager to control her trading strategy x, as it affects the total fund flow M by influencing the clients' expectation about fund's returns.

3.4 Manager's Optimization

By incorporating the total fund flow ([3.7] and [3.9]), the manager's problem in (2.3) is re-written as

$$\max_x \phi \left(sx - \lambda x^2 + \theta \left(\frac{1}{\beta} - \lambda \right) x^2 \right). \tag{3.10}$$

The first two terms represent the fee income from the trading profit and are essentially identical to the trading profit in the original Kyle (1985) model. The third term arises due to the fund inflow and, as equation (3.9) attests, is determined by the quadratic size of risky investments. The optimal trading strategy solves (3.10):

$$x = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1 - \lambda\beta)} s. \tag{3.11}$$

This strategy is consistent with our guess in (3.1) when:

$$\beta = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1 - \lambda\beta)}. \tag{3.12}$$

Solving λ and β in equations (3.3) and (3.12), the trading strategy and the asset price are uniquely determined:

Lemma 3.1. *When the fund information is transparent ($\omega_e = 0$), the manager's trading strategy and the asset price are specified by* (3.1) *and* (3.2) *with*

$$\beta = \sqrt{(1 + 2\theta) \frac{\omega_u}{\omega_s}}, \tag{3.13}$$

and

$$\lambda = \frac{1}{2(1+\theta)}\sqrt{(1+2\theta)\frac{\omega_s}{\omega_u}}. \tag{3.14}$$

In the absence of delegated asset management ($\theta \to 0$), the equilibrium is identical to the original Kyle (1985) model. Hence, β is determined to balance the competing effects of trading intensively. On one hand, the manager seeks to exploit her informational advantage by trading more intensively. On the other hand, it reveals information to the market maker and raises the price impact, cutting into the trading profit.

Delegated asset management ($\theta > 0$) brings about the additional risk-taking incentive, because inflating x benefits the manager by making her appear highly skilled and inducing a larger fund flow (see equations [3.7] and [3.9]). We refer to this incentive as the "showing-off" motive. Since θ governs the flow-performance sensitivity, this motivation becomes stronger when θ is large, shaping equilibrium β into an increasing function of θ.

Equation (3.14) shows that the price impact λ decreases with θ. This is because the information is "exaggerated" in that the order size is too large relative to the true $|s|$ due to the showing-off motive. Thus, the market maker attempts to partially "undo" the manager's information revelation by incorporating less of it into the price, resulting in a weaker price impact.[12]

Our benchmark result in Lemma 3.1 isolates the fundamental mechanism through which delegated asset management enhances the manager's risk taking behavior. The tendency of fund managers to "show off" their skill to attract capital is well discussed in the literature (e.g., Metrick and Yasuda, 2010; Chung, Sensoy, Stern, and Weisbach, 2012; Barber and Yasuda, 2017), raising regulatory concerns that fund managers have incentives to exaggerate the performance or quality of their funds. In our model, however, a skilled manager may also have incentives to conceal her skill from the market maker, as emphasized by Kyle (1985). The next section explores how the fundamental tension between showing off and hiding shapes the equilibrium when the fund

[12] λ represents the price's reaction to the order flow, $x + u$, rather than its reaction to the manager's private information, s. The price impounds s with coefficient $\beta\lambda$, which is monotonically increasing in θ.

information becomes harder to interpret for clients due to opacity.

4 Opaque Funds

This section introduces the fund opacity, $\omega_s > 0$, and demonstrates that belief-driven mltiple equilibria show up regarding the trading strategy of the fund. We discuss economic implications drawn from the equilibrium analyses in Subsection 4.4.

4.1 Delegating to an Opaque Fund

The analysis of an opaque fund ($\omega_s > 0$) largely follows the benchmark case in Section 2. In particular, investors' optimization and the structure of the resulting fund flows are identical to those in (3.6) and (3.7) in the benchmark model. However, due to the noisy revelation of the fund information, investors' conditional expectation about the fund return must be modified as follows:

Lemma 4.1. *Upon observing the fund information, $f = x+e$, investors update their expectation about the trading profit as*

$$\mathrm{E}\left[\pi|f\right] = (1-\lambda\beta)\beta\left(\hat{s}^2 + \hat{\omega}_s\right), \tag{4.1}$$

where the posterior expectation of the (signed) realized skill is

$$\hat{s} = \mathrm{E}[s|f] = \xi(x+e), \tag{4.2}$$

with

$$\xi = \frac{\beta\omega_s}{\beta^2\omega_s + \omega_e}, \tag{4.3}$$

and the posterior of the skill variance is

$$\hat{\omega}_s = \mathrm{Var}[s|f] = \frac{\omega_s\omega_e}{\beta^2\omega_s + \omega_e}. \tag{4.4}$$

Equation (4.2) describes how investors update their belief about the manager's skill using $f = x+e$, where $\hat{s}$ is referred to as the *reputation* in the following discussion. According to equation (4.2), ξ measures how much clients rely on the opaque fund information, determining the sensitivity of the reputation to the fund's trading strategy. Holding β constant, ξ is large when investors' prior is unreliable (ω_s is large) and the fund information is transparent (ω_e is small). More importantly, ξ itself also depends on the trading strategy, β, as a more intensive trading strategy makes the fund information more informative about s.

4.2 Manager's Optimization

Following the same procedures as the transparent benchmark, the manager's optimization problem in (3.10) is re-written as follows:

$$\max_{x} \; sx - \lambda x^2 + \theta(1-\lambda\beta)\beta\xi^2(x^2+\omega_e), \tag{4.5}$$

where we omit the terms unrelated to x. As in the fully transparent benchmark in Section 2, the first two terms of equation (4.5) represent the expected trading profit, and the third term corresponds to her fee income from fund flows, M. Similar to the benchmark case, the total fund flow increases with the manager's risky investment, $|x|$, because she can potentially inflate her reputation $|\hat{s}|$ by placing a large order (either buy or sell). Contrary to the transparent case, however, the reaction of fund flows to the trading strategy is now influenced not only by the flow-performance sensitivity, θ, but also by the clients' updating coefficient, ξ in (4.3). This term kicks in because clients cannot perfectly observe the manager's trading strategy when the fund is opaque. Importantly, clients' estimate of the manager's skill and, therefore, the reaction of fund flows to the risky investment depend on the risky trading strategy itself: the more intensive the trdaing strategy, the more reliable the fund information in estimating the manager's skill. This structure is captured by the dependence of ξ on β in equation (4.3) and leads to the strategic complementarity between the manager's risk taking and her reputation held by clients.

The unique solution to (4.5) (given β) is

$$x = F(\beta)s, \tag{4.6}$$

where

$$F(\beta) \equiv \frac{1}{2\left(\lambda - \theta(1-\lambda\beta)\beta\xi^2\right)}. \tag{4.7}$$

$F(\cdot)$ is a nonlinear in β because λ and ξ are functions of β.

4.3 Equilibrium under Opacity

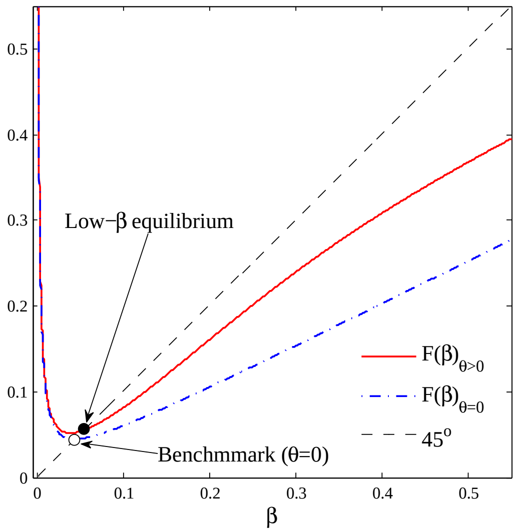
The manager optimally chooses (4.6) given that the market maker and investors believe that she follows strategy (3.1). In equilibrium, their beliefs must be correct, that is, (3.1) and (4.6) must be consistent with each other, requiring the following condition:

$$\beta = F(\beta). \tag{4.8}$$

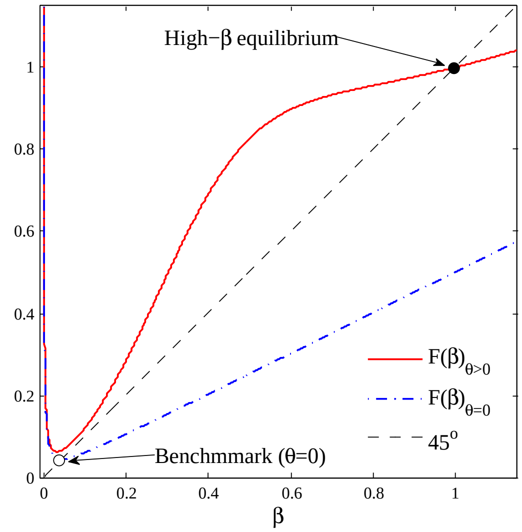
That is, the equilibrium levels of β are determined at the fixed points of $F(\cdot)$. Having identified β, the equilibrium levels of x, p, and m are determined by equations (3.1), (3.2) and (3.6), respectively. In what follows, we first illustrate the determination of equilibrium β through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

Figure 2 illustrates the determination of β for three different levels of fund opacity, ω_e. The intersections of $F(\beta)$ (the solid line) and the 45-degree line yield the equilibrium levels of β. We also plot $F(\beta)$ with $\theta = 0$ (the dash-dotted line) and denote its solution as $\beta_{\theta=0}$ to highlight the original model of Kyle (1985) where fund delegation is irrelevant.

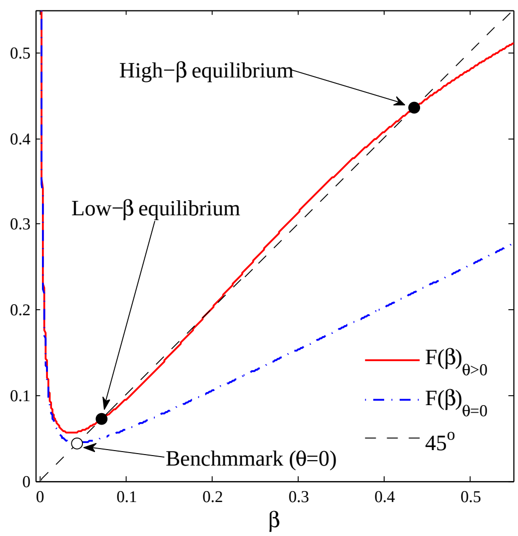
Panel (a) of Figure 2 shows that if ω_e is large, there is a unique "low-β equilibrium" where β is close to $\beta_{\theta=0}$. Panel (b) shows that a large ω_e leads to a unique "high-β equilibrium," in which β is much larger than $\beta_{\theta=0}$. Panel (c) shows that multiple equilibria arise for an intermediate level of ω_e. Specifically, the high-β and the low-β equilibria are stable, as $F(\beta)$ crosses the 45-degree



(a) Small θ ($\theta = 0.2$) $\Rightarrow$ unique low-β equilibrium.



(b) Large θ ($\theta = 0.55$) $\Rightarrow$ unique high-β equilibrium.



(c) Intermediate θ ($\theta = 0.33$) $\Rightarrow$ multiple high-β & low-β equilibria.

Figure 2: Determination of β

Note: The figure plots $F(\beta)$ for different values of θ. The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, and $\omega_u = 0.1$.

line from above, whereas the one with an intermediate β is unstable, as $F(\beta)$ crosses the line from below.[13] In what follows, we focus only on the stable equilibria.

Impact of fund opacity. To see the intuition behind the different equilibrium patterns presented in Figure 2, note that the fund manager faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in Kyle (1985), this motive leads the manager to decrease β. On the other hand, she wishes to trade intensively to signal her skill to her clients, aiming to establish a strong reputation and attract large fund flows. This motive encourages her to increase β. In summary, she faces a tradeoff between "hiding" her skill from the market maker and "showing it off" to her clients. The fund opacity, ω_e, governs this tradeoff by influencing the sensitivity of the manager's reputation held by clients, $\hat{s}$, to her investment behavior, x. In particular, Lemma 4.1 demonstrates that fund flows become less sensitive to x under opacity, as it becomes difficult for clients to estimate the skill based on the fund information. Therefore, fund opacity diminishes the strategic complementarity between the manager's risk taking and the reputation, thereby weakening the showing-off motive.

For an opaque fund with a large ω_e (panel [a]), the hiding motive dominates the showing-off motive, leading to a unique low-β equilibrium. In this equilibrium, clients believe that the manager's trading intensity is relatively low and the fund information is not very informative, making the reputation and fund flows insensitive to $|x|$. Although the manager could inflate her reputation by placing an order larger than what her clients anticipate, she refrains from doing so, because it would cause a large price impact and reduce the fund's investment profits, while fund flows and her fee income would not increase much. Consequently, the equilibrium β is close to $\beta_{\theta=0}$, meaning that it is a

[13] We define an equilibrium to be stable (unstable) if the corresponding β is a stable (unstable) fixed point of $F(\cdot)$, that is, $|F'(\beta)| < 1$ (> 1). A stable equilibrium is robust to a small perturbation to the agents' belief about β, meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents' beliefs.

"Kyle-like equilibrium."

For a transparent fund with a small ω_e (panel [b]), the showing-off motive prevails, resulting in a unique high-β equilibrium. In this equilibrium, clients believe that the manager is trading aggressively. The reputation and fund flows grow sensitive to the manager's trading strategy through the fund information. Then the manager is actually willing to trade aggressively: although it moves the price substantially and undermines investment profits, it helps her build a strong reputation and generates a large fee income from significant capital inflows. Importantly, clients' learning is not manipulated, as their belief about the manager's trading strategy is correct. Nonetheless, the manager places a large order because doing otherwise would induce her clients to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to her clients' belief and to attract large fund flows. To emphasize this mechanism, we call this equilibrium the "flow-driven equilibrium."

For intermediate levels of ω_e (panel [c]), the strategic complementarity between the reputation and the manager's risk taking is neither weak nor strong, and there is no clear dominance between the two motives. Consequently, both the low-β and the high-β equilibria are supported under intermediate fund opacity. Each equilibrium is associated with a self-fulfilling belief of the market maker and investors.

The following proposition and Figure 3 summarize the above discussion and provide an analytical characterization of the equilibria.

Proposition 4.1 (Equilibrium characterization)**.** *There exists a critical value of the fund opacity, $\tilde{\omega}_e$, and two thresholds of the flow-performance sensitivity, θ_L and θ_H ($\geq \theta_L$), such that, $\theta \geq \theta_H$ leads to a unique high-β equilibrium, while $\theta \leq \theta_L$ leads to a unique low-β equilibrium. Furtheremore,*

(i) if $\omega_e \leq \tilde{\omega}_e$, then $\theta_L = \theta_H$, meaning that the equilibrium is unique; and

(ii) if $\omega_e > \tilde{\omega}_e$, then $\theta_H > \theta_L$. In this case, $\theta \in (\theta_L, \theta_H)$ leads to multiple equilibria.

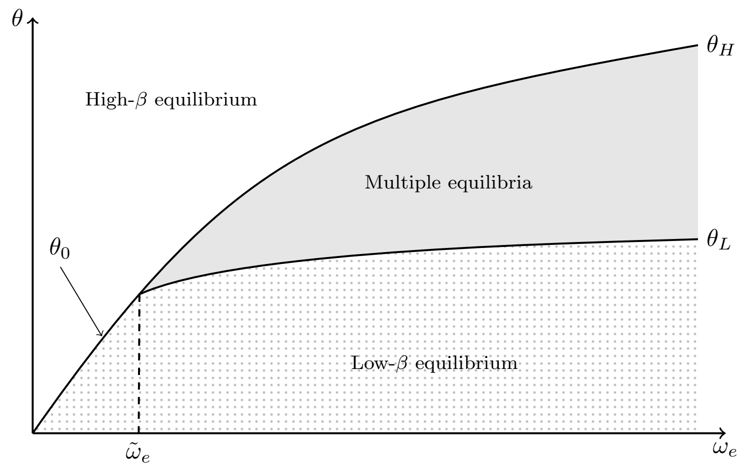


Figure 3: Equilibrium Characterization

Note: This figure illustrates the equilibrium states using the flow-performance sensitivity, θ, and the fund opacity, ω_e.

Impact of flow-performance sensitivity. Proposition 4.1 and Figure 3 indicate that, for a given value of fund opacity ω_e, a weak flow-performance sensitivity supports only the low-β equilibrium, while increases in θ can generate the high-β equilibrium, potentially leading to multiple equilibria.[14] This result is intuitive, as θ, together with ξ, determines the sensitivity of fund flows to the trading strategy, thereby influencing the degree of strategic compelementarity. When θ is small, the reaction fund flows to changes in the expected investment profit becomes weak, and the manager's showing-off motive is less significant. Consequently, the hiding motive prevails, prompting her to trade less aggressively and leading to the low-β equilibrium. Conversely, when θ is large, fund flows become highly sensitive to the fund information, thereby encouraging the manager to show off her skill by trading intensively. This scenario gives birth to the high-β equilibrium.

Overall, the type of equilibrium depends on the combination of θ and ω_e, as illustrated in Figure 3. The tradeoff between the hiding and the showing-

[14] Lemma E.3 in Online Appendix E analytically characterizes the upward-sloping θ_H and θ_L in relation to ω_e, as presented in Figure 3.

off motive tends to balance and generates multiple equilibria when the fund is relatively opaque and fund flows are moderately sensitive to the expected investment performance. Otherwise, one of the conflicting motives dominates the other, making either one of the Kyle-like and flow-driven equilibria a unique equilibrium.

Equilibrium comparison. Comparing the two stable equilibria, does the manager perform better in one of them? What are the implications for market liquidity, asset prices, and learning about asset-selection skills?

Corollary 4.1 (High-β vs. low-β equilibrium)**.** *In the high-β equilibrium, relative to the low-β equilibrium,*

(i) the market liquidity is higher, that is, λ is lower;

(ii) the manager's expected performance given the realized skill, $\mathrm{E}[(\delta - p)x|s] = \frac{\beta\omega_u s^2}{\beta^2\omega_s + \omega_u}$, *is lower;*

(iii) the price informativeness, $\tau \equiv \frac{1}{\mathrm{Var}[\delta|p]} = \frac{\beta^2\omega_s + \omega_u}{\omega_\delta(\beta^2\omega_s + \omega_u) + \omega_u\omega_s}$, *is higher;*

(iv) the price volatility, $\mathrm{Var}[p] = \frac{\beta^2\omega_s^2}{\beta^2\omega_s + \omega_u}$, *is higher; and*

(v) investors' estimate of s is more precise, that is, the posterior variance of s, $\hat{\omega}_s = \frac{\omega_s\omega_u\omega_\delta}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta}$, *is lower.*

Statement (i) of Corollary 4.1 holds because price impact λ is decreasing in β in the equilibrium (see footnote 12). Since the high-β equilibrium emerges only if θ is relatively large, the statement also implies that a higher flow-performance sensitivity leads to higher market liquidity. The intuition is as follows. Relative to the low-β equilibrium, in the high-β equilibrium the manager trades more aggressively, and thus the order flow reflects significant information about s. To incorporate the appropriate amount of information about s into the price, the market maker seeks to offset the exaggerated information in order flow by lowering λ.

The intuition for statement (ii) is closely related to that for statement (i). In the high-β equilibrium, the manager earns a lower trading profit due to a smaller informational advantage over the market maker. This result may appear inconsistent with statement (i) because, ceteris paribus, a lower price impact is typically associated with a higher trading profit. However, the manager's order size $|x|$ in the high-β equilibrium is so large that the *overall* price impact (i.e., λ multiplied by $|x|$) is larger than that in the low-β equilibrium. Consequently, the manager moves the price excessively and earns a lower trading profit in this equilibrium. Due to her aggressive trading, more information about s—which is informative about δ—is incorporated into the price in the high-β equilibrium, supporting statement (iii).

Statement (iv) follows, again, from the manager's aggressive trading in the high-β equilibrium. Specifically, she places a large buy (sell) order when $s > 0$ ($s < 0$), even if $|s|$ is small. This causes a large upward (downward) pressure on p, leading to greater price variability. This result aligns with the insight of Guerrieri and Kondor (2012) that reputation concerns amplify price volatility, although their mechanism differs from ours.[15]

In the high-β equilibrium, the manager reveals more information about s, thereby improving the precision of investors' estimation, as statement (v) demonstrates. This makes fund flows more responsive to skill, as investors can assess it more accurately.

4.4 Implications

This section discusses economic implications drawn from the equilibrium analyses under opacity.

4.4.1 Financial Fragility

The self-fulfilling nature of multiple equilibria suggests that financial markets may become vulnerable to non-fundamental factors. A shift in investor beliefs alone, without any structural changes, could cause the market to transition from the standard Kyle-like equilibrium to the flow-driven equilibrium, characterized by high price volatility and poor trading performance. We interpret

[15] Guerrieri and Kondor (2012) argue that bond price volatility increases because fund managers demand a premium to offset the risk of reputation damage in the event of default.

this phenomenon as a form of *fragility* in financial markets.

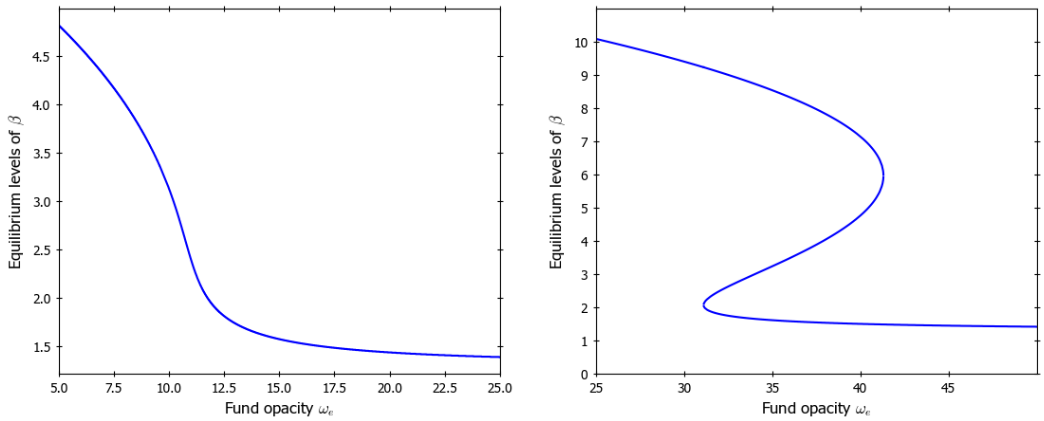
Our model suggests that fragility may occur when the flow-performance sensitivity (θ) is moderately high and the fund is opaque (ω_e). A high flow-performance sensitivity tends to arise when a fund manager covers a large investor base (n), as the total fund flow is increasing in n. More surprisingly, a low fund management fee (ϕ) also leads to the equilibrium multiplicity through heightened θ. This goes counter to the recent debate on fund managers' incentive compensation and advisory contracts, where payment structures that strongly reward performance are often criticized for encouraging excessive risk-taking. As our model implies, such performance-based compensation can actually stabilize the market by eliminating the flow-driven equilibrium and guiding the economy toward a unique low-β equilibrium. This is consistent with the empirical finding by Dass, Massa, and Patgiri (2008), who show that high-powered advisory contracts do not induce investment in bubbly stocks but instead reduce such investments, highlighting the stabilizing effect of incentive compensations on financial markets.

4.4.2 Opacity, Trading Style, and Fund Performance

Furthermore, Proposition 4.1 yields novel implications connecting the fund opacity and its trading styles.

Corollary 4.2. *The trading intensity, β, is monotonically decreasing in the fund opacity, ω_e, both in the high- and low-β equilibria.*

Figure 4 illustrates the result in Corollary 4.2. First, given an intermediate level of θ, opaque funds tend to adopt low-intensity strategies that minimize market impact, prioritizing profits over fund flows, whereas transparent funds engage in high-intensity strategies with flashy risk taking, seeking reputation among investors and large fund flows. This relationship arises because the manager's reputation becomes insensitive to her trading behavior when the fund information is opaque and difficult to interpret for fund clients. The model predicts that these tendencies are more pronounced when the flow-performance sensitivity is weak, as the equilibrium is unique at such levels of



(a) Low flow-performance sensitivity

(b) High flow-performance sensitivity

Figure 4: Fund Opacity and Trading Intensity

Note: these figures are illustrated by seeting $\omega_s = 1.0, \omega_u = 2.0, n = 5, \phi = 0.08$ (the left panel) and $\phi = 0.02$ (the right panel).

θ (panel [a] of Figure 4).

These results are somewhat consistent with common observations in the real market. For example, quantitative hedge funds like Renaissance Technologies and Bridgewater are extremely opaque and highly profit-driven, focusing on consistent performance, with benefits from attracting fund flows being regarded as secondary to profitability (Schwager, 2012). On the other hand, some transparent funds, such as certain thematic ETFs or crypto-focused funds, engage in bold, attention-grabbing strategies to attract investor flows. These funds often emphasize visibility and narrative-driven positioning, which can lead to volatile returns and greater flow sensitivity.

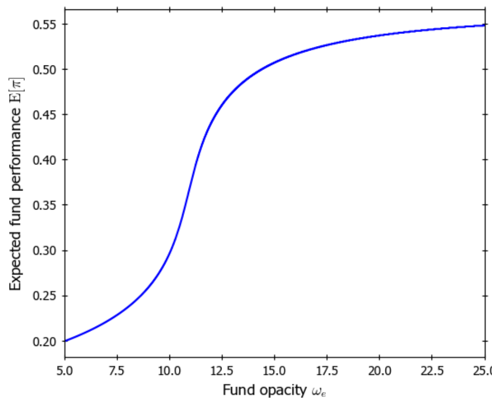
The second implication is drawn from the model's multiple equilibria, as panel (b) in Figure 4 illustrates. Namely, the relation between funds' activities and their transparency may become blurry when the opacity is at an intermediate level. Notably, this indeterminacy may lead to financial fragility. Depending on market beliefs, relatively opaque funds may shift between profit-seeking and flow-driven strategies, creating non-fundamental fluctuations. Our result suggests that even relatively opaque funds may follow the crowd due to market beliefs, hopping between safe and risky strategies.

A notable observation in line with this result is momentum crashes (Daniel and Moskowitz, 2016), where funds engaging in risky and visible momentum trades often switch to safer strategies, generating sharp reversals and amplifying price declines. More generally, the rapid shifts between risk-taking and solid performance-based behavior have contributed to market fragility and volatility, as documented in various studies on economic crises, such as Brunnermeier (2009) on the credit crunch during the Global Financial Crisis in 2007. Importantly, large fundamental shifts that support such transitions are rarely identified (Gennotte and Leland, 1999). Our model does not require such fundamental shocks, as a change in beliefs alone can trigger a shift.

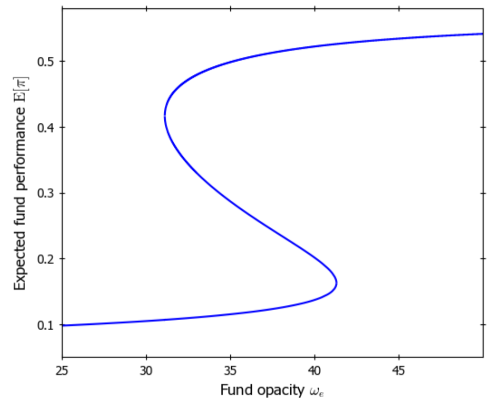
The fund opacity also has an implication for the fund performance. In our model, the manager's expected performance, $\mathrm{E}[\pi]$, is decreasing in her trading intensity, β. This is due to the presence of the showing-off motive: althrough the performance is maximized at the Kyle (1985) benchmark, $\beta = \beta_{\theta=0}$, the manager seeks to trade more aggressively to attract fund flows. It revelas too much private information to the market maker and lowers the fund performance, while the manager is compensated for this loss by attracting more fund flows. The opacity, conversely, diminishes the showing-off motive and induces the manager to take more profit-driven strategies (Corollary 4.2), leading to the following result.

Corollary 4.3. *The expected fund performance is monotonically increasing in the fund opacity both in the high- and low-β equilibria.*

Figure 5 plots the expected profit, $\mathrm{E}[\pi]$, in relation to the fund opacity, ω_e, which traces the reactions of β in Figure 4. Once again, multiple equilibria support somewhat non-trivial impacts of fund opacity to its performance: the market's belief about moderately opaque funds can coordinate on substantially different equilibria, leading similar funds to exhibit divergent performance outcomes. This result implies that empirical studies could find weak or even contradictory correlations between fund transparency and returns.



(a) Low flow-performance sensitivity



(b) High flow-performance sensitivity

Figure 5: Fund Opacity and Expected Performance

Note: The figure plots $F(\beta)$ for different values of θ. The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, and $\omega_u = 0.1$.

4.4.3 Compensation in Fund Management Industry

Our model provides insights into how skills are reflected in compensation in the fund management industry. As shown in statement (v) of Corollary 4.1, fund clients learn s more precisely in the high-β equilibrium than in the low-β equilibrium, as the manager demonstrates her skill by trading more intensively. Consequently, the manager's expected compensation, $\mathrm{E}[\phi y|s]$, become more sensitive to her realized skill, $|s|$. As an illustration, Figure 6 plots the manager's expected compensation in relation to her skill in the low- and high-β equilibria. In the high-β equilibrium, investors' accurate evaluations result in greater pay inequality: low-skilled managers would receive very low compensation, while high-skilled ones are very highly paid. This result corroborates empirical observations, such as those provided by Philippon and Reshef (2012) and Böhm, Metzger, and Strömberg (2018), highlighting the comparatively large wage inequality in the fund management industry. Célérier and Vallée (2019) attribute its source to the scalability of skill, while inequality in our model arises from managers' incentives to attract fund flows through

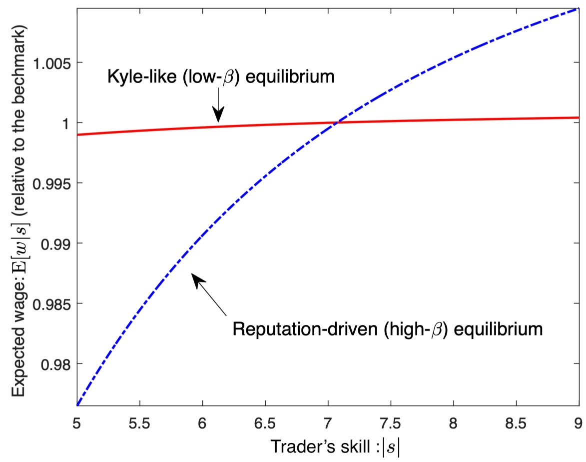


Figure 6: Skill versus Expected Wage

Note: The figure plots the trader's expected wages (relative to the benchmark with $\theta = 0$) at the low-β equilibrium (solid line) and high-β equilibrium (dashed line) for different levels of skill, $|s|$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, $\omega_{u,0} = 0.1$, $\omega_{u,1} = 60$, and $\theta = \frac{\varphi\beta_1}{2} \approx 0.33$ ($\varphi = 0.6$).

establishing high reputation.

4.4.4 Skill and Performance Persistence

Do high-skilled managers consistently perform well? Not always: skilled managers may not necessarily outperform less skilled ones. Consider a highly skilled manager and a relatively less skilled one. Since trading profits are increasing in managers' skill, $|s|$, the former is expected to achieve higher trading profits than the latter *conditional on* both trading at the same intensity, β. However, if their clients believe, for whatever reason, that the skilled manager invest more aggressively than the less skilled one, and managers conform to such beliefs, a high skill level does not translate into performance. This is because the skilled manager, trading at the flow-driven equilibrium, reveals too much private information to the market maker. Hence, compared to the less skilled manager, who plays the Kyle-like equilibrium, the skilled manager may earn lower expected profits. Figure 7 illustrates such a situation. It plots the expected trading profits (performance) at high- and low-β equilibria with

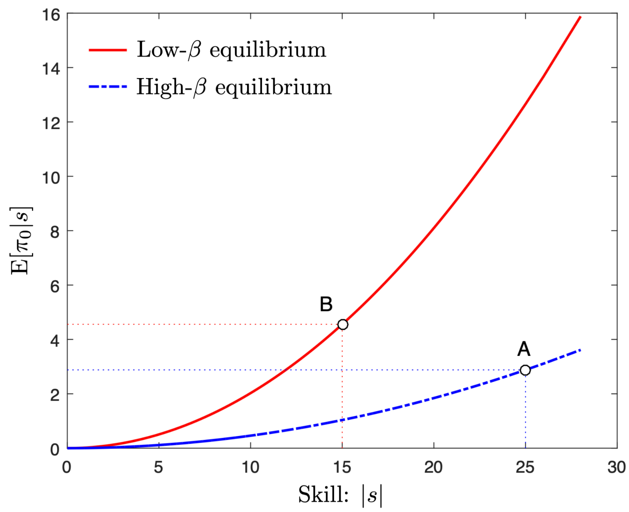


Figure 7: Skill versus Expected Trading Profit

Note: The figure plots the expected trading profit $\mathrm{E}[\pi_0|s]$ at the low-β equilibrium (solid line) and the high-β equilibrium (dashed line) for different levels of skill, $|s|$. The parameter values used in the figure are $\omega_s = 50$, $\omega_\delta = 1$, $\omega_u = 0.1$, and $\theta = \frac{\varphi\beta_1}{2} \approx 0.33$ ($\varphi = 0.6$).

different levels of skill, suggesting that a low-skilled manager in the low-β equilibrium (point B) outperforms a high-skill manager in the high-β equilibrium (point A).

The empirical literature (e.g., Jensen, 1967; Busse, Goyal, and Wahal, 2010) finds that asset managers do not outperform passive benchmarks on average and, in cross-section, those who outperform cannot do so consistently. While the literature attributes the lack of persistent performance to the lack of skill, our result proposes an alternative explanation. Several studies reconcile the existence of skill with the lack of persistent performance. Berk and Green (2004), for example, argue that skill-based alphas eventually disappear due to competition between investors and decreasing returns to scale at the fund level.[16] Our model contributes to the understanding of this matter by showing that the lack of interplay between manager performance and skill could be the

[16]Pastor, Stambaugh, and Taylor (2015) document increases in skill in finance and attribute the lack of corresponding increases in performance to decreasing returns to scale at the industry level.

result of self-fulfilling multiple equilibria.

5 Strategic Opacity

This section analyzes endogenous opacity by allowing the manager to control ω_e. It also endogenizes the negative reaction of the fund clientele to opacity, capturing the cost of making the fund opaque for the manager. We find that she strategically chooses opacity as a commitment device to restrict herself from taking excessive risk.

5.1 Setup

We extend the model to a two-stage setting ($t = 0, 1$), assuming that the baseline model analyzed in the previous sections takes place at $t = 1$.

At the begenning of $t = 0$, the fund manager chooses the fund opacity, ω_e. Subsequently, each investor decides whether to participate in the fund by paying cost $C(\omega_e)$ with $C(0) = 0$ and $C'(\cdot) > 0$, which we justify below. The investors' participation decisions collectively determine the size of clientele, n. At the begenning of $t = 1$, the manager realizes her skill, s, and the model evolves as specified in Section 2. In the end of $t = 1$, each client observes the realized asset's payoff and its price, (δ, p), and receives the payment from the fund according to the manager's report about fund proceeds, denoted as $\tilde{y}$.

Misreporting and verification. The fund manager has an incentive to make a false report about the fund proceeds. Namely, by reporting $\tilde{y}$, she pays out $(1-\phi)\tilde{y}$ to clients and captures $y - (1-\phi)\tilde{y}$ on her own, suggesting that reporting $\tilde{y} = 0$ would be optimal.[17] To counteract this intentional misreporting, we assume that a client pays cost C and can verify x *ex-post* in the end of $t = 1$. Together with the market data (δ, p), she can verify $y = (\delta - p)x + M$, thereby

[17] Technically, the manager earns an infinite payment by reporting $\tilde{y} = -\infty$, while we assume that clients must receive non-negative payoffs from the fund (i.e., they do not pay additional money), thereby imposing $\tilde{y} \geq 0$.

disproving $\tilde{y} \neq y$.[18] Otherwise, the manager can intentionally misreport $\tilde{y} = 0$, as there are levels of e and u that are consistent with the report, i.e., she claims that the investment was $\tilde{x} = -M/(\delta - p)$, while the noise components in the fund information and order flow were $\tilde{e} = f - \tilde{x}$ and $\tilde{u} = (x+u) - \tilde{x}$, making f and p consistent with $\tilde{y} = 0$. The verification cost is increasing in the degree of fund opacity, $C'(\omega_e) > 0$, indicating that clients need more intensive analyses to extract information about the fund's strategy when the fund becomes more opaque.

Sunspot shock. To formalize the optimization problems in $t = 0$ in the rational expectations framework, we introduce an equilibrium-selection device in cases of multiple equilibria. In particular, following the most common approach, we assume that market participants condition their actions on the realization of an exogenous *sunspot* shock, characterized by the binary random variable $z \in \{0, 1\}$.[19] Namely, whenever parameters admit multiple equilibria, market participants play the high-β equilibrium if a spot appears on the sun ($z = 1$), whereas the low-β equilibrium is realized if no spots are observed ($z = 0$). Accordingly, the high-β equilibrium is realized with probability $\rho_H \equiv \Pr(z = 1)$, while the complementary probability is denoted as $\rho_L = 1 - \rho_H$. We assume that z is realized at the beginning of the trading stage in $t = 1$. In what follows, the trading intensity in the high-β and low-β equilibria are denoted as $\beta = \beta_H$ and β_L, respectively. Also, for consistency, we define $\rho_j = 1$ for $j = H, L$ when the equilibrium with $\beta = \beta_j$ is unique.

Optimization. For each level of β, equation (3.4) leads to the following expected payoff to individual investors:

$$V_I(\beta) \equiv \mathrm{E}\left[\frac{1-\phi}{n}\mathrm{E}[\pi|f] - \phi\frac{\theta}{n}\mathrm{E}[\pi|f]\right] = \frac{1-\phi}{n^2}\frac{\beta\omega_s\omega_u}{\beta^2\omega_u + \omega_s}, \tag{5.1}$$

[18] We implicitly assume that the manager is penalized and incurs a negative large utility if her intentional misreporting is detected.

[19] The sunspot equilibrium is proposed by Diamond and Dybvig (1983) as an equilibrium selection device in the context of bunk runs and formalized in the equilibrium analyses by the subsequent studies, such as Cooper and Ross (1998).

where $\beta = \beta_H$ or β_L. The first term in the middle of (5.1) represents the share of the trading profit allocated to each client, and the second term is the cost assosiated with delegation, ϕm, in the equilibrium. The last equality arises from the expected trading profit, $\mathrm{E}[\pi] = \frac{\beta\omega_s\omega_u}{\beta^2\omega_u+\omega_s}$. Therefore, an investor is willing to participate in the fund and pays the verification cost if, and only if,

$$\bar{V}_I \equiv \sum_{j=H,L} \rho_j V_I(\beta_j) \geq C(\omega_e), \tag{5.2}$$

where $\bar{V}_I$ represents the investor's *ex-ante* expected payoff, incorporating multiple equilibria. As formalized below, the size of clientele (n) increases as long as inequality (5.2) holds. It determines the equilibrium n and influences the manager's compensation through the flow-performance sensitivity, $\theta = \frac{n-1}{n}\frac{1-\phi}{\phi}$.

Conversely, for each level of β, the manager's expected payoff is expressed as

$$V_M(\beta) \equiv \mathrm{E}\left[\mathrm{E}\left[\phi\pi + (1-\phi)\frac{n-1}{n}\mathrm{E}[\pi|f]\right]\right] = \phi(1+\theta)\frac{\beta\omega_s\omega_u}{\beta^2\omega_u+\omega_s}. \tag{5.3}$$

Incorporating the reaction of her clients (n), as well as the possibility of sunspot-driven multiple equilibria, the manager controls fund opacity (ω_e) to maximize the following expected payoff:

$$\max_{\omega_e} \sum_{j=H,L} \rho_j V_M(\beta_j). \tag{5.4}$$

5.2 Equilibrium

We first solve for the participation decisions of clients and pin down the equilibrium size of clientale, n, as a function of fund opacity, ω_e.

Fund participation.

1. The expected profit from delegating fund is monotonically decreasing in n. Therefore, the equilibrium size of clientele is determined by the

break-even condition: $\bar{V}_I = C(\omega_e)$.

2. There is a unique $n = n(\omega_e)$ that satisfies the break-even condition, and it serves as the equilibrium size of clientele. Moreover, $n(\omega_e)$ depends on opacity: there are two competing effect: the commitment channel and the costly-verification channel. The latter is obvious.

   (a) The former is interesting and related to XXX.

   The fund investor size is decreasing function of ω_e as long as the marginal verification cost is sufficiently large. Surprisingly, when the verification cost is small, $n(\omega_e)$ is increasing in ω_e, suggesting that opaque funds attract more investors than transparent funds.

3. As long as the commitment channel is dominant, the manager has an incentive to make her fund opaque: it forces her to limit the trading intensity and increases her payoffs both from the trading and the fund flows. However, as the opacity increases, the costly-verification channel kicks in, and the size of investors start decreasing.

4. So, when ω_e is low, the benefit from opacity is large: the trading profit is increasing not only due to the reduced β but also due to the commitment channel which increases n. However, as ω_e becomes large, the costly-verification channel becomes significant, thereby reducing the clients size. Although the profit is still increasing, this decline becomes the cost for the manager.

5.3 Equilibrium

We first solve the problems in $t = 1$ and obtain the salary, w, which is shown to be a quadratic function of the perceived skill, $|\hat{s}|$. Subsequently, we solve the problems in $t = 0$, which are similar to those of Section 2.

5.3.1 Period 1

In $t = 1$, each trader has no reputation concerns. Hence, she simply chooses x_1 to maximize π_1, resulting in the standard Kyle (1985) equilibrium.

Lemma 5.1 (Equilibrium in $t=1$). *In $t=1$, each trader's trading strategy is characterized by $x_1=\beta_1 s$, and the market maker sets the execution price according to $p_1=\hat{\delta}_1+\lambda_1\left(x_1+u_1\right)$, where $\beta_1 \equiv \sqrt{\frac{\omega_{u,1}}{\omega_s}}$ and $\lambda_1 \equiv \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}$.*

Lemma 5.1 yields the trader's profit, π_1. Then, the *recruiter's* expectation about π_1 determines the trader's salary when she meets the recruiter.

Lemma 5.2 (Trader's salary). *Let $\hat{s}$ and $\hat{\omega}_s$ be the posterior mean and variance of s from the recruiter's point of view. The salary he offers to the trader is*

$$w=\frac{\beta_1}{2}\left(\hat{s}^2+\hat{\omega}_s\right). \tag{5.5}$$

Lemma 5.2 gives the competitive wage, w, as a quadratic function of $\hat{s}$, providing a microfoundation for the reputation benefit assumed in Section 2.

5.3.2 Period 0

In $t=0$, the trader knows that she will potentially obtain w in the form of (5.5). Hence, plugging (5.5) into (**??**), her objective function in $t=0$ is

$$\mathrm{E}\left[\pi_0+\frac{\varphi\beta_1}{2}\left(\hat{s}^2+\hat{\omega}_s\right)+(1-\varphi)\pi_1\middle|s\right]. \tag{5.6}$$

Since π_1 and $\hat{\omega}_s$ are independent of x_0, her optimization problem in $t=0$ is characterized by (5.7) in the following proposition.

Proposition 5.1 (Endogenous reputation). *In $t=0$, each trader's maximization problem is identical to (**??**) of Section 2, that is,*

$$\max_{x_0}\mathrm{E}\left[\pi_0+\theta\hat{s}^2\middle|s\right], \tag{5.7}$$

where parameter θ for reputation concerns is given by

$$\theta\equiv\frac{\varphi}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}. \tag{5.8}$$

Proposition 5.1 provides a microfoundation to the quadratic form of the reputation benefit in Section 2. Moreover, θ is endogeneized as a composite parameter (5.8). Not surprisingly, it is increasing in φ. It decreases with the trader's skill potential, ω_s, suggesting that a high-potential trader is less concerned about her reputation. This is due to the *future* trading intensity β_1 and the price impact λ_1: a high-potential trader is likely to possess a high skill, which induces a large price impact in the future as well. Since it is difficult for the trader to exploit her skill in $t = 1$, the competitive wage becomes less responsive to $|s|$, weakening the reputation concern.

As problem (5.7) is effectively the same as (**??**), the equilibrium outcomes resemble Proposition 4.1. However, unlike Proposition 4.1, the impact of the skill potential ω_s on the equilibrium characterization is generalized due to the dependence of θ on ω_s. This is summarized by the following proposition, which is graphically presented in Figure 8.

Proposition 5.2 (Equilibrium in $t = 0$)**.** *The equilibrium at $t = 0$ is characterized as follows.*

1. *If φ is large enough, there are two thresholds for the skill potential, denoted as $\underline{\omega}_s > 0$ and $\bar{\omega}_s$ ($\geq \underline{\omega}_s$), such that*

    (i) for $\omega_s \geq \bar{\omega}_s$, the low-$\beta$ equilibrium is uniquely realized;

    (ii) for $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, both the high- and low-β equilibria arise;

    (iii) for $\omega_s \leq \underline{\omega}_s$, the high-$\beta$ equilibrium is uniquely realized.

2. *If φ is small enough, there is a unique equilibrium. For small ω_s, the high-β equilibrium is realized, while for large ω_s, the low-β equilibrium is realized.*

Since the trader's reputation concern θ is endogeneized as (5.8), which is a decreasing function of ω_s, Figure 8 is obtained by overlaying (5.8) on Figure 3. The first and the second parts of Proposition 5.2 correspond to panels (a) and (b) of Figure 8, respectively. Proposition 5.2 states that multiple equilibria may arise only when labor mobility is sufficiently high (φ is large), i.e., only when the traders have a large reputation concern, θ.[20] Since the case in panel

[20]Equation (E.15) in Online Appendix E provides the definition of the boundary value of φ that classifies cases 1 and 2 in Proposition 5.2.

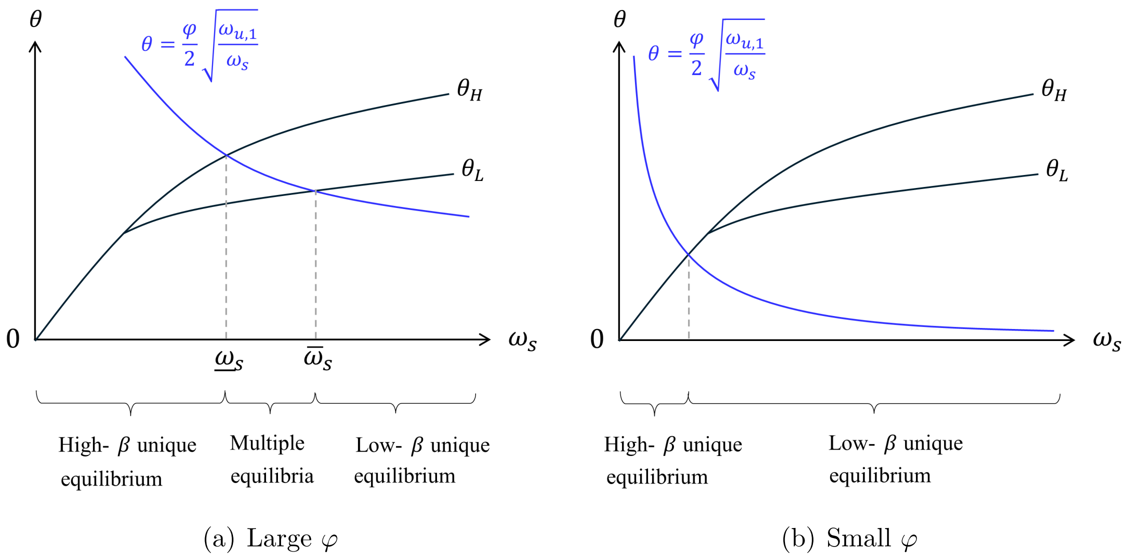


Figure 8: Equilibrium Characterization with Endogenous θ

Note: These figures illustrate the equilibrium states using the reputation concern, θ, and the skill potential, ω_s. The curves represented by θ_H and θ_L are identical to those in Figure 3 and Proposition 4.1, while θ is endogenous and decreasing in ω_s as equation (5.8) suggests.

(a) yields richer implications, while panel (b) is qualitatively similar to Kyle (1985), we focus on the case in panel (a), which arises when φ is relatively large.

Panel (a) of Figure 8 shows that, as a result of endogeneizing θ as a function of ω_s, the conditions characterizing the equilibrium types simplify to a single condition solely on ω_s: a large ω_s yields a unique low-β equilibrium, an intermediate ω generates multiple equilibria, and a small ω_s leads to a unique high-β equilibrium.

The intuition is as follows. When the skill potential is high ($\omega_s \geq \bar{\omega}_s$), the trader is more likely to have a large informational advantage over the market maker. This leads to a strong price impact, making it costly to inflate the trading strategy to receive high reputation. Consequently, she is more inclined to adopt profit-seeking strategies, and the Kyle-like equilibrium is uniquely realized. As the skill potential decreases, however, the trader's informational advantage and the price impact tend to diminish, making it easier to prioritize her reputation concerns. Therefore, for intermediate levels of $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, the reputation-driven equilibrium arises and coexists with the Kyle-like equilibrium. If the skill potential declines further ($\omega_s \leq \underline{\omega}_s$), the reputation becomes the dominant factor, and the Kyle-like equilibrium disappears.

5.4 Setup

Before the trading stage in $t = 0$, each trader makes a costly investment to improve her skill potential. This investment could include pursuing an advanced degree in finance, studying for professional exams, or engaging in specialized training to expand her expertise. The trader selects her skill potential, denoted as ω_s^*, by paying the skill-acquisition cost, $C(\omega_s^*) = c\omega_s^*$ with a positive marginal cost $c > 0$.[21] We assume that ω_s^* is not observable to the market makers and the recruiter. Instead, they form a belief about it, denoted by ω_s, and the equilibrium in the trading stage is determined by ω_s following Proposition 5.2. On the equilibrium path, the belief must be consistent with the actual skill potential, $\omega_s^* = \omega_s$. However, at the skill-acquisition stage, the trader cannot influence ω_s, as she does not have a commitment device.[22]

Expected utility of a trader. Given the realized equilibrium in the first trading stage (i.e., β_0), the *ex-ante* expected utility of a trader is characterized by the following lemma:

Lemma 5.3. *For each β_0, the expected utility of a trader at the skill-acquisition stage is given by*

$$V(\omega_s^*, \omega_s, \beta_0) = v(\omega_s, \beta_0)\omega_s^* + \bar{v}, \tag{5.9}$$

[21]The following analysis does not hinge on the functional form of the cost, as long as it is weakly increasing in ω_s^*.

[22]This setting follows the literature on private information acquisition with an unobservable quality (precision) of the private signal. See, for example, Xiong and Yang (2023).

where $\bar{v}$ summarizes constant terms unrelated to ω_s^*,[23] *and $v(\omega_s, \beta_0)$ is*

$$v(\omega_s, \beta_0) \equiv (1 - \lambda_0 \beta_0)\beta_0 + (1 - \psi)\frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}} + \theta\xi^2 \left[\gamma^2 \beta_0^2 + 2\gamma(1-\gamma)\beta_0 + (1-\gamma)^2\right]. \tag{5.10}$$

Equation (5.9) indicates that, for each unit of the selected skill potential (ω_s^*), the trader earns the marginal utility of $v(\omega_s, \beta_0)$. The marginal utility in equation (5.10), in turn, consists of three factors. The first two terms represent the maximized trading profit in $t = 0$ and that in $t = 1$ when she does not meet the recruiter. In the first trading round, the equilibrium profit margin of the skill potential is $(1 - \lambda_0\beta_0)\beta_0$, while the profit margin in the second trading round is identical to that in the standard Kyle (1985) model.[24] The third term is the reaction of the reputation benefit to an increase in the selected skill potential, which emanates from the reputation concerns, θ, and the magnitude of the recruiter's belief updating (see equations [**??**] and [5.5]).

Importantly, $v(\omega_s, \beta_0)$ is characterized by the market's belief about the trader's skill potential (ω_s), as ω_s^* is unobservable to the market. The trader cannot influence ω_s by controlling ω_s^* due to limited commitment. Consequently, the optimality condition for the skill acquisition is characterized by ω_s, as the following discussion elaborates.

Sunspot shock. To formalize the skill-acquisitiom problem based on Lemma 5.3, we introduce an equilibrium-selection device. In particular, following the most common approach, we assume that market participants condition their actions on the realization of an exogenous *sunspot* shock, characterized by the binary random variable $z \in \{0, 1\}$.[25] Namely, when $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, they play

[23] The formal expression is given by $\bar{v} = \frac{\psi}{4\lambda_1}\left[\xi^2\gamma^2\omega_{u,0} + \xi^2(1-\gamma)^2\omega_\delta + \omega_{u,0}\frac{\gamma\xi}{\beta_0}\right]$

[24] In the standard Kyle (1985) model, the optimal trading intensity is $\beta_0 = \frac{1}{2\lambda_0}$, so that the equilibrium profit margin of the information advantage (i.e., the skill in our model) is directly captured by $\frac{\beta_0}{2}$. Our model does not have this feature, as the trading intensity is set by incorporating both the price impact and the reputation concern.

[25] The sunspot equilibrium is proposed by Diamond and Dybvig (1983) as an equilibrium selection device in the context of bunk runs and formalized in the equilibrium analyses by

the high-β equilibrium if a spot appears on the sun ($z = 1$), whereas the low-β equilibrium is realized if no spots are observed ($z = 0$). Accordingly, the high-β equilibrium is realized with probability $\Pr(z = 1) = \rho$, where ρ denotes the exogenous probability of a sunspot. We assume that z is realized at the beginning of the trading stage in $t = 0$, meaning that a trader acquires her skill potential before z is realized. In what follows, the high-β and low-β equilibria are denoted as $\beta_0 = \beta_H$ and β_L, respectively.

By incorporating the equilibrium multiplicity stipulated in Proposition 5.2, the trader's *ex-ante* expected utility from acquiring ω_s^* in equation (5.9) is rewritten as follows:[26]

$$\mathrm{E}_\beta[v(\omega_s, \beta_0)]\omega_s^* = \begin{cases} v(\omega_s, \beta_L)\omega_s^* & \text{if } \omega_s \geq \bar{\omega}_s, \\ \left[\rho v(\omega_s, \beta_H) + (1-\rho) v(\omega_s, \beta_L)\right]\omega_s^* & \text{if } \omega_s \in (\underline{\omega}_s, \bar{\omega}_s), \\ v(\omega_s, \beta_H)\omega_s^* & \text{if } \omega_s \leq \underline{\omega}_s. \end{cases} \tag{5.11}$$

E_β denotes the expectation over $\beta_0 = \beta_H$ and β_L in cases of multiple equilibria and incorporates the sunspot shock with $\rho = \Pr(z = 1) = \Pr(\beta_0 = \beta_H)$.

5.5 Equilibrium

Prior to the trading game, each trader solves the following problem by controlling her skill potential:

$$\max_{\omega_s^*} \mathrm{E}_\beta[v(\omega_s, \beta_0)]\omega_s^* - C(\omega_s^*), \tag{5.12}$$

where $\mathrm{E}_\beta[v(\omega_s, \beta_0)]\omega_s^*$ is given by equation (5.11). As the trader cannot control ω_s, she takes the marginal benefit in (5.11) as given.

In the equilibrium, the marginal benefit balances the marginal cost of skill acquisition, as equation (5.13) below illustrates. Otherwise, the trader has an incentive to increase or decrease ω_s^*. Together with the belief-consistency

the subsequent studies, such as Cooper and Ross (1998).

[26]We omit the constant term, $\bar{v}$, in the following discussion, as it does not affect the optimal skill potential.

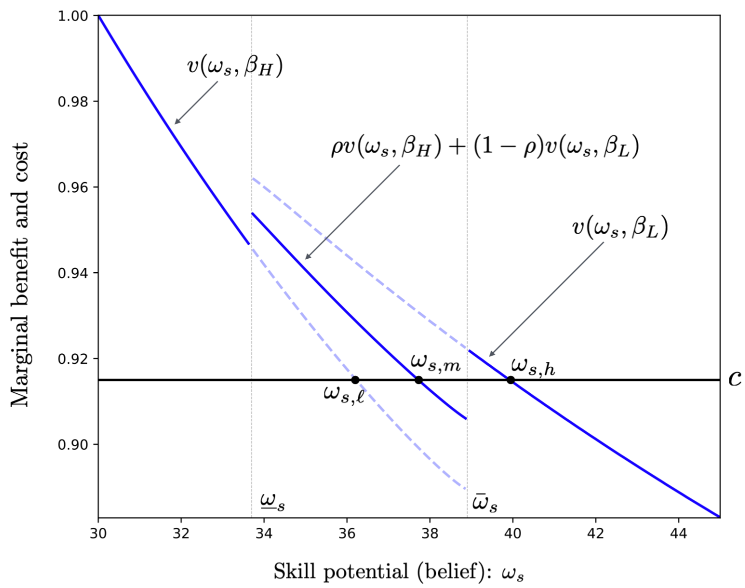


Figure 9: Marginal Benefit and Cost of Skill Acquisition

Note: The figure plots the marginal benefit of skill acquisition, given by $\mathrm{E}_\beta[v(\omega_s, \beta_0)]$, and its marginal cost c against the belief about the skill potential, ω_s. The parameter values are $\omega_\delta = 1$, $\omega_{u,0} = 0.1$, $\omega_{u,1} = 60$, $\theta = \frac{\varphi\beta_1}{2} \approx 0.33$ ($\varphi = 0.6$), and $\rho = 0.5$.

condition, the optimal skill potential ω_s^* is derived as solutions to the following optimality condition with respect to ω_s:

$$\mathrm{E}_\beta[v(\omega_s, \beta_0)] = c. \tag{5.13}$$

Figure 9 plots the marginal benefit (the LHS of [5.13]) in relation to the market's belief about the skill potential, ω_s. Although it is orthogonal to the actual skill potential, ω_s and ω_s^* on the equilibrium path are determined by the intersections of the marginal benefit and the marginal cost curves in Figure 9.

The marginal benefit of skill acquisition globally diminishes as ω_s increases. This result is not obvious because there are two channels through which ω_s affects its marginal benefit. In the first channel, when the market maker believes that the trader has acquired a high ω_s, the price impact grows, and the trading intensity diminishes in both trading rounds at $t = 0$ and $t = 1$, thereby reducing the benefit of increasing ω_s^*. Conversely, in the second channel, when ω_s is high, the recruiter faces higher prior uncertainty in estimating the realized skill ($|s|$), and the wage becomes more reliant on the price information (p_0) and the asset payoff in $t = 0$ (δ_0). Hence, the wage becomes more sensitive to the trader's skill, making it more valuable for the trader to acquire a higher skill potential ω_s^*. Although the second channel encourages the trader to increase ω_s^*, the negative effect in the first channel is dominant, forming the downward-sloping marginal benefit curve in Figure 9.

Notably, the marginal benefit curve in the skill-acquisition stage has discontinuities due to the multiple equilibria in the trading stage. When the (belief of) skill potential is sufficiently high or low ($\omega_s \geq \bar{\omega}_s$ or $\omega_s \leq \underline{\omega}_s$), β_0 in the trading stage is uniquely determined. However, when the skill potential turns to intermediate levels, the trader starts incorporating the possibility of multiple equilibria in the trading stage, generating jumps in the marginal benefit of skill acquisition. By focusing on the intermediate levels of $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$, we obtain the following result.

Lemma 5.4. *With ω_s being fixed, the marginal benefit of increasing the skill potential is smaller in the high-β equilibrium than in the low-β equilibrium. Therefore, the following inequalities hold, and the marginal benefit curve has discontinuities at $\omega_s = \underline{\omega}_s$ and $\omega_s = \bar{\omega}_s$:*

$$v(\omega_s, \beta_H) < \mathrm{E}_\beta[v(\omega_s, \beta_0)] < v(\omega_s, \beta_L). \tag{5.14}$$

Intuitively, the trader in the high-β equilibrium excessively reveals information to the market maker by trading aggressively. Although the recruiter estimates the trader's skill more precisely and increases the wage's sensitivity to the skill potential, it is dominated by the negative impact of a reduction in the trading profit margin. Therefore, the marginal benefit of increasing the skill potential becomes smaller in the high-β equilibrium than in the low-β equilibrium, leading to inequalities (5.14). In what follows, we elaborate on the discontinuities and derive their implications for skill acquisition.

5.5.1 Middling Skill Acquisition

We first propose "middling" skill acquisition due to the multiple trading-stage equilibria by focusing on $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$. We refer to it as middling because the acquired skill level is either higher or lower than what she would acquire if she knew which type of equilibrium would be realized. On the equilibrium path, both the selected skill potential (ω_s^*) and the market's belief about it (ω_s) fall in this region if the marginal skill-acquisition cost is intermediate. In this region, the trading stage involves multiple equilibria (i.e., $\beta_0 = \beta_H$ or β_L), whereas the trader at the skill-acquisition stage cannot influence the equilibrium selection. Consequently, she determines her skill potential by anticipating the sunspot shock, which leads to the skill acquisition as follows.

Proposition 5.3 (Middling skill acquisition)**.** *The equilibrium skill potential in the presence of multiple trading-stage equilibria is*

(i) lower than the skill potential that she would acquire in the low-β equilibrium; and

(ii) higher than the skill potential that she would acquire in the high-β equilibrium.

Figure 9 illustrates a numerical example; $\omega_{s,m}$ is the ex-ante optimal skill potential, while the ex-post equilibrium is either $\omega_{s,\ell}$ or $\omega_{s,h}$, depending on the equilibrium realization in the trading stage.

The trader invests in skill potential incorporating the possibility of both high- and low-β equilibria. At the trading stage, however, the acquired skill potential turns out to be insufficient if the low-β equilibrium is realized, as its optimal play involves the profit-driven strategy and requires a relatively high asset-selection skill (i.e., $\omega_{s,h}$ is the *ex-post* equilibrium). Conversely, the acquired skill potential is too high if the high-β equilibrium is realized, as the trader takes the reputation-driven strategy that does not require a high skill potential (i.e., $\omega_{s,\ell}$ is the *ex-post* equilibrium).

This result indicates that individuals or firms may invest in skills that seem oddly intermediate or cautious—neither fully specialized nor overly basic. For example, it explains why some traders and finance professionals choose to pursue broader business qualifications instead of more specialized academic paths, allowing them to adapt to a wide range of market situations. Traders also learn intermediate-level coding (e.g., Python) for algorithmic trading or data analysis but stop short of deep computer science expertise. A similar trend can be observed with respect to CFA Levels and quantitative skills among hedge fund managers. These skill acquisitions would be difficult to rationalize in an economy with a single equilibrium but become logical under the equilibrium multiplicity and uncertainty regarding equilibrium realization.

5.5.2 Fragile Skill Acquisition and Skill Diversity

By expanding our scope to the entire range of skill potential (ω_s), the discontinuities in Lemma 5.4 imply that the skill-acquisition problem in (5.12) may have multiple solutions.[27]

For example, when the marginal cost of skill acquisition is moderately small, as Figure 9 illustrates, both the high skill potential ($\omega_{s,h}$) and the intermediate skill potential ($\omega_{s,m}$) become optimal for the trader. Corresponding to $\omega_{s,h}$, the equilibrium in the trading stage is unique with $\beta_0 = \beta_L$, where the trader engages in solid profit-driven trading. Conversely, the trading stage with skill potential $\omega_{s,m}$ involves multiple equilibria, and the selected skill potential is middling, as Proposition 5.3 demonstrates. The symmetric argument holds when the marginal cost is relatively high: the intermediate skill potential and the low skill potential show up as multiple optima. This result can explain at least three realistic phenomena without relying on ad-hoc assumptions on the skill-acquisition environment.

Firstly, the coexistence of traders with different skill levels arises not from heterogeneity in skill-acquisition problems (e.g., different marginal costs), but from the multiple trading-stage equilibria and the possibility of mixed strategies in the skill acquisition stage. This reflects a realistic scenario where, due

[27]Although the equilibrium multiplicity in the trading stage (i.e., β_0) is determined by the level of ω_s, as Proposition 5.2 stipulates, it is orthogonal to the existence of multiple solutions to the optimal skill acquisition in equation (5.13).

to advancements in the internet and social media (e.g., online webinars via Zoom and academic courses uploaded on YouTube), opportunities for skill acquisition have become widely and uniformly accessible to all traders, reducing barriers to learning. Despite the increased homogeneity in skill-acquisition costs, we continue to observe heterogeneity in skill levels in the finance industry. Our model offers an explanation for this observation, showing how diverse skill levels can emerge among market participants, without the need to assume inherent differences in their skill acquisition costs.

Secondly, it also captures the possibility of widespread surges in skill acquisition without assuming special conditions for the skill-acquisition environment. For example, we can explain traders and fund managers in competitive environments (e.g., Wall Street) may collectively strive to acquire high-level skills, such as pursuing advanced degrees, driven by the mixed-strategy equilibrium. This surge in skill acquisition could occur even in the absence of significant changes in external factors, and without the need for heterogeneity in skill-acquisition costs, reflecting industry-wide trends in upskilling and degree inflation, as documented by Philippon and Reshef (2012) and Fuller and Raman (2017).

Finally, discontinuities in the marginal benefit curve add another layer of insight into the effect of the skill-acquisition cost on the equilibrium skill potential. Namely, discontinuities introduce the possibility of significant jumps in the optimal skill level triggered by a small change in the marginal skill-acquisition cost. For instance, when marginal technological advancements slightly reduce the marginal cost of learning, the result could be a sizable upward shift in skill acquisition. Conversely, even a small increase in the cost of learning might cause a large decline in the equilibrium skill level. This discontinuity-driven sensitivity can help explain the historical observations stylized by Philippon and Reshef (2012) that certain external changes lead to abrupt shifts in the distribution of skill levels in the finance industry.

6 Conclusion

This paper studies a model à la Kyle (1985) where traders are privately informed about their skill, i.e., the trader-specific ability to identify potentially mispriced assets. The traders care not only about trading profits but also the recruiter's perception of their skill. The model has two stable self-fulfilling equilibria, which we interpret as fragility in financial markets. One of the equilibria is similar to that of Kyle (1985) where reputation is irrelevant. In the other, "reputation-driven" equilibrium, the traders trade more aggressively for fear of being seen as unskilled by the recruiter, causing higher price volatility, higher market liquidity, and lower trading profits. Our model predicts that fragility may arise when poaching is (moderately) active in the finance industry and when there are high demands for financial skill. The model yields implications connecting traders' skills, their trading styles, and financial fragility. Moreover, consistent with empirical observations, our model predicts large pay inequality in the finance industry and reconciles skill in active management and the lack of performance persistence.

As a direction for future research, it would be interesting to analyze various types of wage structures in financial markets, ranging from back-end to front-end compensations. The implications of the forward-looking trading motives introduced in our model can be applied to investigate which types of wage structures and employment systems can mitigate financial market fragility while also improving the welfare of market participants.

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Online Appendix

A Proof of Lemma ??

From (3.1) and (3.2), p is observationally equivalent to $\frac{p-\hat{\delta}}{\lambda} = \beta s + u \equiv z_p$, which is the sum of two normally distributed variables from the market maker's perspective. Since $\delta \sim N(\hat{\delta} + s, \omega_\delta)$ is equivalently presented as $\delta = \hat{\delta} + s + \epsilon$ with $\epsilon \sim N(0, \omega_\delta)$, δ is observationally equivalent to $\delta - \hat{\delta} = s + \epsilon \equiv z_\delta$, which is also the sum of normal variables. Thus, given $Z = [z_p,\ z_\delta]^{\mathrm{T}}$, the market maker's posterior mean of s is

$$\mathrm{E}\,[s|Z] = \underbrace{\mathrm{E}[s]}_{=0} + \Sigma_{sZ}\Sigma_Z^{-1}(Z - \underbrace{E[Z]}_{=[0\ \ 0]^{\mathrm{T}}}), \tag{A.1}$$

where

$$\Sigma_{sZ} = \begin{bmatrix} \beta\omega_s, & \omega_s \end{bmatrix} \text{ and } \Sigma_Z = \begin{bmatrix} \beta^2\omega_s + \omega_u, & \beta\omega_s \\ \beta\omega_s, & \omega_s + \omega_\delta \end{bmatrix}. \tag{A.2}$$

Since

$$\begin{aligned} \det[\Sigma_Z] &= (\beta^2\omega_s + \omega_u)(\omega_s + \omega_\delta) - \beta^2\omega_s^2 \\ &= \omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta, \end{aligned} \tag{A.3}$$

it holds that

$$\begin{aligned} \mathrm{E}\,[s|Z] &= \begin{bmatrix} \beta\omega_s, & \omega_s \end{bmatrix} \frac{1}{\det[\Sigma_Z]} \begin{bmatrix} \omega_s + \omega_\delta, & -\beta\omega_s \\ -\beta\omega_s, & \beta^2\omega_s + \omega_u \end{bmatrix} \begin{bmatrix} z_p \\ z_\delta \end{bmatrix} \\ &= \frac{\omega_s}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left(\beta\omega_\delta z_p + \omega_u z_\delta\right) \\ &= \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left(\frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u} z_p + \left(1 - \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}\right) z_\delta \right) \\ &= \xi\left(\gamma z_p + (1-\gamma) z_\delta\right), \end{aligned} \tag{A.4}$$

where

$$\xi \equiv \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \tag{A.5}$$

and

$$\gamma \equiv \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}. \tag{A.6}$$

Re-writing z_p and z_δ in (A.4) in terms of p and δ, we obtain the required result. $\square$

B Proof of Lemma ??

Let us present $\delta \sim N(\hat{\delta} + s, \omega_\delta)$ equivalently as $\delta = \hat{\delta} + s + \epsilon$ with $\epsilon \sim N(0, \omega_\delta)$. Then, squaring (**??**), we have

$$\begin{aligned}
\hat{s}^2 &= \xi^2 \left(\gamma^2 x^2 + 2\gamma x \left(\gamma u + (1-\gamma)(\delta - \hat{\delta}) \right) + \left(\gamma u + (1-\gamma)(\delta - \hat{\delta}) \right)^2 \right) \\
&= \xi^2 \left(\begin{array}{c} \gamma^2 x^2 + 2\gamma^2 x u + 2\gamma(1-\gamma)(s+\epsilon)x \\ +\gamma^2 u^2 + 2\gamma(1-\gamma)(s+\epsilon)u + (1-\gamma)^2(s+\epsilon)^2 \end{array} \right).
\end{aligned} \tag{B.1}$$

Taking the expectation of (B.1) from the trader's perspective,

$$\mathrm{E}[\hat{s}^2|s] = \xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx + A \right), \tag{B.2}$$

where

$$A \equiv \gamma^2\omega_u + (1-\gamma)^2(s^2 + \omega_\delta) \tag{B.3}$$

is the term that the trader is unable to influence by her choice of x. We can omit A in the maximization problem below. Using (B.2), the trader's problem (**??**) is rewritten as

$$\begin{aligned}
&\max_x \; \mathrm{E}\left[(\delta - p)x|s\right] + \theta\mathrm{E}[\hat{s}^2|s] \\
\Rightarrow \; &\max_x \; \mathrm{E}\left[\left(\hat{\delta} + s + \epsilon - \left(\hat{\delta} + \lambda(x+u) \right) \right) x \middle| s \right] + \theta\mathrm{E}[\hat{s}^2|s] \\
\Rightarrow &\max_x \; sx - \lambda x^2 + \theta\xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx \right). \quad \square
\end{aligned} \tag{B.4}$$

C Proof of Lemma 5.1

At the beginning of $t = 1$, the trader does not have reputation concerns anymore. Her problem is just to choose x_1 to maximize the expected trading profit, that is,

$$\max_{x_1 \in (-\infty,\infty)} \mathrm{E}[(\delta_1 - p_1)x_1|s]. \tag{C.1}$$

We conjecture and later verify that the trader's equilibrium order in $t = 1$ is

$$x_1 = \beta_1 s, \tag{C.2}$$

where $\beta_1 > 0$ is a constant to be determined. Given (C.2), the market maker sets the price

$$\begin{aligned}
p_1 =& \mathrm{E}\left[\delta_1 | x_1 + u_1\right] \\
=& \mathrm{E}[\mathrm{E}[\delta_1|x_1 + u_1, s]|x_1 + u_1] \\
=& \mathrm{E}[\hat{\delta}_1 + s|x_1 + u_1] \\
=& \hat{\delta}_1 + \frac{\mathrm{Cov}[s, x_1 + u_1]}{\mathrm{Var}[x_1 + u_1]}\left(x_1 + u_1 - \mathrm{E}[x_1 + u_1]\right) \\
=& \hat{\delta}_1 + \frac{\mathrm{Cov}[s, \beta_1 s + u_1]}{\mathrm{Var}[\beta_1 s + u_1]}\left(x_1 + u_1 - \mathrm{E}[\beta_1 s + u_1]\right) \\
=& \hat{\delta}_1 + \lambda_1\left(x_1 + u_1\right),
\end{aligned} \tag{C.3}$$

where

$$\lambda_1 \equiv \frac{\beta_1 \omega_s}{\beta_1^2 \omega_s + \omega_{u,1}}. \tag{C.4}$$

Given (C.3), the trader's problem (C.1) is rewritten as

$$\max_{x_1} \ \mathrm{E}\left[\left(\delta_1 - \left(\hat{\delta}_1 + \lambda_1(x_1 + u_1)\right)\right) x_1 \middle| s\right] \ \Rightarrow \ \max_{x_1} \ s x_1 - \lambda_1 x_1^2. \tag{C.5}$$

The FOC for x_1 is $s - 2\lambda_1 x_1 = 0$, which determines the trader's optimal action:

$$x_1 = \frac{1}{2\lambda_1} s. \tag{C.6}$$

For the other agents' belief (C.2) to be consistent with (C.6), we need

$$\beta_1 = \frac{1}{2\lambda_1} \quad \Longleftrightarrow \quad \beta_1 = \sqrt{\frac{\omega_{u,1}}{\omega_s}}, \tag{C.7}$$

which implies

$$\lambda_1 = \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}. \quad \square \tag{C.8}$$

D Proof of Lemma 5.2

Let us present $\delta_1 \sim N(\hat{\delta}_1 + s, \omega_\delta)$ equivalently as $\delta_1 = \hat{\delta}_1 + s + \epsilon_1$ with $\epsilon_1 \sim N(0, \omega_\delta)$. Then the trader's trading profit in $t = 1$ is

$$\begin{aligned}
\pi_1 &= (\delta_1 - p_1)x_1 \\
&= \left(\hat{\delta}_1 + s + \epsilon_1 - \left(\hat{\delta}_1 + \lambda_1\left(\beta_1 s + u_1\right)\right)\right)\beta_1 s \\
&= \left(\hat{\delta}_1 + s + \epsilon_1 - \left(\hat{\delta}_1 + \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}\left(\sqrt{\frac{\omega_{u,1}}{\omega_s}}s + u_1\right)\right)\right)\sqrt{\frac{\omega_{u,1}}{\omega_s}}s \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}s^2 + \left(\sqrt{\frac{\omega_{u,1}}{\omega_s}}\epsilon_1 - \frac{u_1}{2}\right)s. \qquad \text{(D.1)}
\end{aligned}$$

Taking the expectation of (D.1) from the recruiter's perspective, we obtain the trader's salary:

$$\begin{aligned}
w &= \mathrm{E}[\pi_1 | p_0, \delta_0] \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\mathrm{E}[s^2 | p_0, \delta_0] \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\left(\mathrm{E}[s|p_0, \delta_0]^2 + \mathrm{Var}[s|p_0, \delta_0]\right) \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\left(\hat{s}^2 + \hat{\omega}_s\right), \qquad \text{(D.2)}
\end{aligned}$$

where $\hat{\omega}_s$ is the posterior variance of s, which is a constant. $\square$

E Proof of Propositions 4.1 and 5.2

In the following analyses, denote

$$a = \frac{\omega_{u,0}}{\omega_s},\ b = \frac{\omega_{u,0}}{\omega_\delta}, \tag{E.1}$$

so that we rewrite equation (4.8) as

$$\beta = F(\beta) \equiv \frac{1 + 2\theta \left(\frac{1}{a+b+\beta^2}\right)^2 b\beta}{2\left(\frac{\beta}{\beta^2+a} - \theta\left(\frac{\beta}{a+b+\beta^2}\right)^2\right)}. \tag{E.2}$$

By rearranging (E.2), the equilibrium β solves the following condition.

$$2\theta\beta = J(\beta^2), \tag{E.3}$$

where, by denoting $\beta^2 = y$,

$$J(y) = \frac{(y-a)(y+b+a)^2}{(y+a)(y+b)}. \tag{E.4}$$

The first-order derivative of the RHS of (E.3) with respect to β is

$$K^{(1)}(\beta) \equiv 2\beta J^{(1)}(\beta^2), \tag{E.5}$$

with

$$J^{(1)}(y) \equiv \frac{N(y)}{(y+a)^2(y+b)^2}, \tag{E.6}$$

where the numerator is

$$N(y) = y^4 + 2(a+b)y^3 + n_2 y^2 + n_1 y + n_0$$

with coefficients being

$$n_2 = 2a^2 + 6ab + b^2,$$

$$n_1 = 2a\left[3b^2 + 3ab + a^2\right],$$

$$n_0 = a\left(a^2 + ab + 2b^2\right)(a+b).$$

From (E.4) and (E.6), $y = a$ is the unique solution to $J(y) = 0$ in $y > 0$, and $J(y)$ is increasing in y for $y \geq a$. Hence, there is no equilibrium in $y < a$, and we focus on $y \geq a$.

The second-order derivative of the RHS of (E.3) with respect to β is expressed as

$$K^{(2)}(y) \equiv 2\left(J^{(1)}(y) + 2yJ^{(2)}(y)\right), \tag{E.7}$$

where $J^{(2)}$ denotes the second-order derivative of J with respect to y. $K^{(1)}$ and $K^{(2)}$ have the following properties:

Lemma E.1. *(i) $K^{(1)}(\beta)$ takes a U-shaped curve in relation to β.*
(ii) $K^{(2)}(a) < 0$ and $K^{(2)}(\infty) > 0$. There exists a unique solution to $K^{(2)}(y) = 0$.

Proof. Note that point (i) directly follows from point (ii). By denoting that $N^{(1)}(y) = \frac{dN(y)}{dy}$, it holds that

$$J^{(1)}(y)+2yJ^{(2)}(y) = J^{(1)}(y)\frac{-7y^2 - 3(a+b)y + ab}{(y+a)(y+b)} + 2y\frac{\left(y^2 + (a+b)y + ab\right)N^{(1)}(y)}{(y+a)^3(y+b)^3}. \tag{E.8}$$

Evaluating at $y = a$,

$$\begin{aligned} J^{(1)}(a) + 2aJ^{(2)}(a) &= -J^{(1)}(a)\frac{5a+b}{a+b} + \frac{N^{(1)}(y)}{2a(a+b)^2} \\ &\propto -2a(a+b)(5a+b)J^{(1)}(a) + N^{(1)}(a) \\ &= -4a^3 - b(4+b)a - b^3 \\ &< 0. \end{aligned}$$

Therefore, $K^{(2)}(a) < 0$ holds.

The numerator of (E.8) is summarized as

$$T(y) = -\left[y^4 + 2(a+b)y^3 + n_2y^2 + n_1y + n_0\right]\left[7y^2 + 3(a+b)y - ab\right]$$
$$+ 2y\left(y^2 + (a+b)y + ab\right)\left[4y^3 + 6(a+b)y^2 + 2n_2y + n_1\right].$$

The above argument suggests that $T(a) < 0$.

Using Mathematica, we summarize T into the following polynomial:

$$T(y) = \sum_{k=0}^{6} t_k y^k, \tag{E.9}$$

where the coefficients are

$$t_0 = a^2b(a+b)\left(a^2 + ab + 2b^2\right) > 0,$$

$$t_1 = -3a\left(a^4 + (a+b)(a-b)^2b + b^4\right) < 0,$$

$$t_2 = -3a\left(3a^3 + 4a^2b + ab^2 + 5b^3\right) < 0,$$

$$t_3 = -8a^3 - 8a^2b - 9ab^2 + b^3,$$

$$t_4 = 3b\,(a+b) > 0,$$

$$t_5 = 3(a+b) > 0,$$

and $t_6 = 1$. As $t_6 > 0$, $T(\infty) = \infty > 0$. Together with $T(a) < 0$, the intermediate value theorem implies that $T(y) = 0$ has at least one solution in $y > a$.

Suppose that y_{min} is one of the solutions to $T(y) = 0$. Consider the first-order derivative of T evaluated at $y = y_{min}$ and multiplied y_{min}:

$$\mathcal{T}(y_{min}) = y_{min}\frac{dT(y_{min})}{dy} = \sum_{k=1}^{6} k t_k y_{min}^k.$$

Applying $T(y_{min}) = 0$ and eliminating the terms related to t_3,

$$\mathcal{T}(y_{min}) = -3t_0 - 2t_1y_{min} - t_2y_{min}^2 + t_4y_{min}^4 + 2t_5y_{min}^5 + 3t_6y_{min}^6,$$

According to the signs of t_k and $y_{min} > a$, it holds that

$$
\begin{aligned}
\mathcal{T}(y_{min}) &> -3t_0 - 2t_1 a - t_2 a^2 + t_4 a^4 + 2t_5 a^5 + 3t_6 a^6 \\
&= 3a^2 \left(6a^4 + 8a^3 b + b^4 + (a^2 - b^2)^2\right) \\
&> 0.
\end{aligned}
$$

Therefore, if y_{min} is a solution to $T(y) = 0$, then the slope of $T(y)$ at this point must be positive. Together with $T(a) < 0 < T(\infty)$, y_{min} is the unique solution to $T(y) = 0$, and $T(y) > 0 \Leftrightarrow y > y_{min}$.[28] The argument above implies that $K^{(1)}(\beta)$ takes a U-shaped curve with respect to β, and the minimum point is given by $\beta_{min} \equiv \sqrt{y_{min}}$. Note that $\tilde{\beta}$ in footnote **??** is given by $\tilde{\beta} = \beta_{min}$. □

The RHS of (E.3) is monotonically increasing in β, and Lemma E.1 attests that it is concave in $y \leq y_{min}$, while it becomes convex in $y > y_{min}$. In other words, the slope of the curve is the least steep at $y = y_{min}$, and it is given by

$$K_{min} = 2\beta_{min} J^{(1)}(y_{min}). \tag{E.10}$$

Note that K_{min} is represented only by parameters, a, b, and does not involve θ.

Consider the following condition to characterize equilibrium:

$$J(y_{min}) > K_{min}\beta_{min}. \tag{E.11}$$

Then the following result holds:

Lemma E.2. *(i) If inequality (E.11) holds, then there exist θ_H and θ_L with $\theta_H > \theta_L$. If, and only if, $\theta \in \Theta \equiv (\theta_L, \theta_H)$, then the equilibrium condition (E.3) has three solutions.*
(ii) If inequality (E.11) is violated, then $\theta_H = \theta_L$, and the equilibrium condition (E.3) has a unique solution. The solution is $\beta > \beta_{min}$ ($\beta < \beta_{min}$) if $\theta > \theta_H$

[28] Suppose that $T(y) = 0$ has more than two solutions. For any solutions, denoted as y_0 and y_1, the mean-value theorem implies that there exists at least one $y_2 \in [y_0, y_1]$ such that $\frac{dT(y_2)}{dy} = 0$. This contradicts to the strict positivity or negativity of $\mathcal{T}$.

($\theta < \theta_H$).

Proof. Consider the plot of the curve, $J(\beta^2)$, and the line, $2\theta\beta$, against β. With the curve being fixed, consider a line that goes through the origin and is tangent to the curve. The tangent point is characterized by β that satisfies

$$J(\beta^2) = K^{(1)}(\beta^2)\beta,$$

where the LHS represents the y-coordinate of the curve at β, and the RHS represents that of the line that goes through the origin. Note that their slopes must be identical and $K^{(1)}(\beta^2)$. Applying (E.5), the above condition is rewritten as

$$J(\beta^2) = 2\beta^2 J^{(1)}(\beta^2).$$

Hence, denoting $y = \beta^2$, the tangent point is characterized by the solution to

$$0 = G(y) = 2yJ^{(1)}(y) - J(y).$$

Note that we focus on $y > \sqrt{a}$, and thus y and β have one-to-one mapping.

Function $G(y)$ has the following first-order derivative:

$$\begin{aligned} G^{(1)}(y) &= 2\left(yJ^{(2)}(y) + J^{(1)}(y)\right) - J^{(1)}(y) \\ &= J^{(1)}(y) + 2yJ^{(2)}(y) \\ &= \frac{K^{(2)}(y)}{2}, \end{aligned}$$

where we apply the definition of $K^{(2)}$ in (E.7). Lemma E.1 has established that $K^{(2)}(a) < 0$ and $K^{(2)}(\infty) > 0$. Also, there is a unique $y = y_{min}$ such that $K^{(2)}(y_{min}) = 0$ and $K^{(2)}(y_{min}) > 0 \Leftrightarrow y > y_{min}$. Hence, $G(y)$ takes a U-shaped curve with respect to y and $y = y_{min}$ is the minimizer of the curve.

With $G(y)$ being U-shaped, $G(y) = 0$ has two solutions if and only if its minimum point dips below the x-axis, namely,

$$2y_{min}J^{(1)}(y_{min}) - J(y_{min}) < 0.$$

Applying the definition of K_{min} in (E.10), the above inequality is rewritten as condition (E.11).

When condition (E.11) holds and $G(y) = 0$ has two solutions, we denote them as y_0 and $y_1 (> y_0)$. The corresponding slopes of these tangent lines are $K^{(1)}(y_0)$ and $K^{(1)}(y_1)$, respectively. Then the original line, $2\theta\beta$, and the curve, $J(\beta^2)$, have three intersections if and only if

$$\frac{K^{(1)}(y_1)}{2} \equiv \theta_L < \theta < \theta_H \equiv \frac{K^{(1)}(y_0)}{2}. \tag{E.12}$$

This result concludes the proof of statement (i) of the lemma.

Conversely, if condition (E.11) does not hold, no tangent line that goes through the origin is obtained: function $J(\beta^2)$ becomes the least steep at $\beta_{min} = \sqrt{y_{min}}$, and the tangent line is specified by $K_{min}\beta - K_{min}\beta_{min} + J(\beta_{min}^2)$, while the intercept of this line is negative, as condition (E.11) is violated. Hence, the line, $2\theta\beta$, and the curve, $J(\beta^2)$ have a unique intersection. This intersection occurs in $\beta > \beta_{min}$ if, and only if, $\theta > \theta'_H \equiv \frac{J(\beta_{min}^2)}{2\beta_{min}}$.

To conclude statement (ii), we must show that θ_H and θ_L in statement (i) converges to θ'_H in statement (ii) at the threshold of these two cases so that we re-define it as $\theta_H = \theta_L = \theta'_H$. This result is shown in Lemma E.3 below. $\square$

Finally, consider the comparative statics with respect to ω_s. The following result holds:

Lemma E.3. *(i) $J(y_{min}) - K_{min}\beta_{min}$ is increasing in ω_s. Hence, there exists a unique $\tilde{\omega}_s > 0$, such that, (E.11) holds with equality, and inequality (E.11) holds if and only if $\omega_s > \tilde{\omega}_s$.*
(ii) When condition (E.11) holds, both θ_H and θ_L increase as ω_s increases. Interval Θ expands, as the following inequalities hold:

$$0 < \frac{d\theta_L}{d\omega_s} < \frac{d\theta_H}{d\omega_s}.$$

(iii) At $\omega_s = \tilde{\omega}_s$, it holds that $\theta_H = \theta_L = \theta'_H$, while at the limit of $\omega_s \to 0$, $\theta'_H \to 0$.

(iv) There exists a unique threshold for φ^*, such that, $\varphi \geq \varphi^*(< \varphi^*)$ leads to case 1 (case 2) in Proposition 5.2.

Proof of Statements (i) and (ii). Note that we are interested in the reaction of the following function when $a = \frac{\omega_{u,0}}{\omega_s}$ changes and, hereafter, we make the dependence of functions on a explicit:

$$2\theta_k = K^{(1)}(y_j, a) = 2\sqrt{y_j} J^{(1)}(y_j, a),$$

where $y_j = y_0, y_1$ denote the solutions to $G(y, a) = 0$, if any, and $k = L, H$ are appropriately determined following (E.12). According to the definition of G, the following partial derivative holds.

$$\frac{\partial G(y,a)}{\partial a} = 2y \frac{\partial J^{(1)}(y,a)}{\partial a} - \frac{\partial J(y,a)}{\partial a}.$$

The first term is summarized by

$$\begin{aligned}
\frac{\partial J^{(1)}(y,a)}{\partial a} &\propto (y+a)\frac{\partial N(y,a)}{\partial a} - 2N(y,a) \\
&= (y+a)\left[2y^3 + n_{2,a}y^2 + n_{1,a}y + n_{0,a}\right] - \left[y^4 + 2(a+b)y^3 + n_2 y^2 + n_1 y + n_0\right] \\
&= \sum_{k=0}^{4} A_k y^k
\end{aligned}$$

where $A_4 = 1,\ A_3 = 4(a+b)$,

$$A_2 = 2a^2 + 5b^2 + 2(3ab + a^2) + 2a\left[3b + 2a\right],$$

$$A_1 = 2a^2\left[3b + 2a\right]\left(a^2 + ab + 2b^2\right)(2a+b) + a(2a+b)(a+b),$$

$$\begin{aligned}
A_0 &= a\left[(2a+b)\left(a^2 + ab + 2b^2 + a(a+b)\right) - \left(a^2 + ab + 2b^2\right)(a+b)\right] \\
&> (a(a+b))^2.
\end{aligned}$$

Hence, $\frac{\partial J^{(1)}(y,a)}{\partial a} > 0$. The second term is

$$(y+b)\frac{\partial J(y,a)}{\partial a} = -2\frac{(y+b+a)\left(y(a+b)+a^2\right)}{(y+a)^2}$$
$$< 0. \tag{E.13}$$

Therefore, $\frac{\partial G}{\partial a} > 0$. As the previous proof has established that $G(y,a)$ takes a U-shaped curve, $\frac{\partial G}{\partial a} > 0$ implies that its minimum point is monotonically decreasing in ω_s, concluding the first statement of point (i) in the lemma.

As $a \to 0$, it holds that $y_{min} \to 0$ and

$$\lim_{a\to 0} G(y_{min}, a) = \lim_{a\to 0}[2y_{min} - (y_{min} + b)] = -b < 0.$$

Also

$$\lim_{a\to\infty} G(y_{min}, a) = +\infty.$$

Therefore, together with

$$\frac{dG(y_{min}, a)}{da} = \frac{dy_{min}}{da}\frac{\partial G}{\partial y} + \frac{\partial G}{\partial a} > 0,$$

the above limits imply the existence of $\bar{a}$, such that, condition (E.11) holds if and only if $a < \bar{a}$. Given $\omega_{u,0}$, this result can be translated into ω_s so that $\tilde{\omega}_s = \frac{\omega_{u,0}}{\bar{a}}$. Noting that $a < \bar{a} \Leftrightarrow \omega_s > \tilde{\omega}_s$, we conclude the second half of point (i) in the lemma.

Suppose that condition (E.11) holds, meaning that y_0 and y_1 exist. For $j \in 0, 1$, $\frac{\partial G}{\partial a} > 0$ implies that $\frac{dy_0}{da} > 0$ and $\frac{dy_1}{da} < 0$. Also, the implicit function theorem leads to

$$\frac{dy_j}{da} = -\frac{1}{G^{(1)}(y_j, a)}\frac{\partial G(y_j, a)}{\partial a}.$$

By taking the total derivative of θ_k,

$$\begin{aligned}
\frac{d\theta_k}{da} &= \frac{1}{2}\frac{1}{\sqrt{y_j}}\frac{dy_j}{da}G^{(1)}(y_j,a) + \sqrt{y_j}\frac{\partial J^{(1)}(y_j,a)}{\partial a} \\
&= -\frac{1}{2}\frac{1}{\sqrt{y_j}}\left(2y_j\frac{\partial J^{(1)}(y_j,a)}{\partial a} - \frac{\partial J(y_j,a)}{\partial a}\right) + \sqrt{y_j}\frac{\partial J^{(1)}(y_j,a)}{\partial a} \\
&= \frac{1}{2}\frac{1}{\sqrt{y_j}}\frac{\partial J(y_j,a)}{\partial a} \qquad \text{(E.14)} \\
&< 0,
\end{aligned}$$

where the last inequality follows from (E.13). Therefore, we conclude that the boundary slopes, θ_H and θ_L are both increasing in ω_s.

Consider the magnitude of the reactions. By plugging (E.13) into (E.14),

$$\frac{d\theta_k}{da}| \propto \frac{\sqrt{y_j}}{(y_j+a)^2}\left(1+\frac{a}{y_j+b}\right)\left(1+\frac{1}{y_j}\frac{a^2}{a+b}\right).$$

It is straightforward that the last terms are all decreasing in y_j. Since we compare $y_0 < y_1$ with a being fixed, we obtain $\frac{d\theta_H}{d\omega_s} > \frac{d\theta_L}{d\omega_s}$, thereby concluding the proof.

Proof of Statements (iii) and (iv). Note that $\lim_{\omega_s\to\tilde{\omega}_s} y_j = y_{min}$, and $J(y_{min}) = K_{min}\beta_{min} = 2\beta_{min}J^{(1)}(y_{min})$ holds. Therefore, at this limit,

$$\theta'_H = \frac{J(\beta^2_{min})}{2\beta_{min}} = \beta_{min}J^{(1)}(\beta^2_{min}) = \frac{K_{min}}{2} = \theta_k,$$

where the last equality holds because of (E.5). This concludes the proof of statement (iii).

Finally, consider the limit at $\omega_s \to 0$, which implies $a \to \infty$ (given $\omega_{u,0}$). Since $y_{min} \searrow a$, we can denote it as $y_{min} = a + \epsilon$, where $\epsilon > 0$ is a small positive and $\lim_{\omega_s\to\tilde{\omega}_s}\epsilon = 0$. With this notation, function J is rewritten at the limit as

$$\lim_{\omega_s\to\tilde{\omega}_s} J(y_{min},a) = \lim_{a\to\infty,\epsilon\to 0}\frac{\epsilon(2a+b+\epsilon)}{(2a+\epsilon)(a+b+\epsilon)} = 0.$$

Since $\theta'_H = \frac{J(y_{min})}{2\sqrt{y_{min}}}$, we conclude that $\lim_{\omega_s \to \tilde{\omega}_s} \theta'_H = 0$.

As for statement (iv), case 1 occurs if, and only if, the curve $\theta = \frac{\varphi}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}$ exceeds θ_H at $\omega_s = \tilde{\omega}_s$. The above analyses have already established that $\theta_H = \frac{K^{(1)(y_{min})}}{2}$, where y_{min} is implicitly characterized by parameters a and b, we can define the threshold φ^* as

$$\varphi^* \equiv K^{(1)(y_{min})}\sqrt{\frac{\omega_{u,0}}{\bar{a}\omega_{u,1}}}. \tag{E.15}$$

□

Note that Proposition 4.1 follows from points (i) and (ii) of Lemma E.3, whereas Proposition 5.2 is shown by putting monotonically decreasing θ in equation (5.8) together with points (i) and (ii).

F Proof of Lemma 5.4 and Proposition 5.3

The optimal trading strategy satisfies the following condition.

$$\therefore (1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1}\xi^2(\gamma^2\beta_0 + \gamma(1-\gamma)) = 0. \tag{F.1}$$

The marginal benefit of increasing s^* is given by

$$\mathrm{E}_\beta[v(\omega_s, \beta_0)] = \left[(1 - \lambda_0\beta_0) + \frac{\psi}{4\lambda_1}\xi^2(\gamma^2\beta_0 + 2\gamma(1-\gamma))\right]\beta_0 + \frac{\psi}{4\lambda_1}\xi^2(1-\gamma)^2 + (1-\psi)\frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}. \tag{F.2}$$

Then, it holds that

$$\begin{aligned}\frac{\partial \mathrm{E}_\beta[v(\omega_s, \beta_0)]}{\partial \beta_0} &= \left[(1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1}\xi^2(\gamma^2\beta_0 + \gamma(1-\gamma))\right] \\ &\quad + \left[-\beta_0\frac{\partial \lambda_0}{\partial \beta_0} + \frac{\psi}{4\lambda_1}\frac{\partial}{\partial \beta_0}\left[\xi^2(\gamma^2\beta_0 + 2\gamma(1-\gamma))\right]\right]\beta_0 + \frac{\psi}{4\lambda_1}\frac{\partial}{\partial \beta_0}\left(\xi^2(1-\gamma)^2\right).\end{aligned}$$

Note that the first term is zero due to the envelope theorem using (F.1). By using the notations introduced in equations (E.1), the remaining terms are summarized as follows.

$$-\beta_0 \frac{\partial}{\partial \beta_0} \frac{\beta_0}{\beta_0^2 + a} + \frac{\psi}{4\lambda_1} \left[\beta_0^2 \frac{\partial}{\partial \beta_0} \left(\frac{\beta_0}{a + b + \beta_0^2} \right)^2 + 2\beta_0 \frac{\partial}{\partial \beta} \left(\frac{1}{a + b + \beta_0^2} \right)^2 b\beta_0 + \frac{\partial}{\partial \beta} \left(\frac{b}{a + b + \beta_0^2} \right)^2 \right]. \tag{F.3}$$

From the proof in Appendix E, it holds that $y > a$ in the equilibrium. Therefore, the first term is negative. As for the second term, it holds that

$$\begin{aligned} \text{Second terms of (F.3)} &= 2\beta_0^3 \left(\frac{a + b - \beta_0^2}{(a + b + \beta_0^2)^3} \right) + 2 \frac{a + b - 3\beta_0^2}{(a + b + \beta_0^2)^3} b\beta_0 - 4b^2 \frac{\beta_0}{(a + b + \beta_0^2)^3} \\ &\sim -y^2 + y(a + b - 3b) + b(a - b). \end{aligned}$$

Since $y > a$, the quadratic equation above is negative for all $y > a$ if and only if it is negative at $y = a$. This is true because

$$-a^2 + a(a - 2b) + b(a - b) = -ab - b^2 < 0.$$

Therefore, the marginal benefit in (F.2) is decreasing in β_0, concluding the proof of Lemma 5.4. Proposition 5.3 directly follows from Lemma 5.4 and equation (5.13).